Supplementary Material: When Agreement Hides Error

Anonymous CVPR submission

Paper ID *****

This supplement is organized by provenance, theory, protocols, complete results, negative results, external scope and reproduction. Historical results remain unchanged unless an explicitly documented semantic or numerical correction is stated.

1. Provenance and Study Status

1.1. Frozen final iteration and information boundaries

Exactly three new scientific analyses were permitted: all ordered cross-architecture phase contrasts, acquired-only neural influence, and a fixed-context perturbation test of Proposition 1. Original papers and inputs were preserved before new inference. This is a frozen retrospective iteration on reused cohorts, not preregistration. The optional unused-cohort branch is skipped because no authenticated untouched cohort was established. Administrative submission metadata are outside this scientific iteration.

The SAINT coordinate amendment was recorded after interface inspection but before any new inference: both new phases use physical $z/(\text{source depth} - 1)$. Legacy coordinates are preserved for reproduction. New FP32 finite-response probes also differ from historical mixed precision. These are explicit implementation contracts, not favorable-result substitutions. Raw arrays and the initial/amended protocol hashes are retained.

1.2. Provenance, Implementation Correction, and Claim Levels

This revision is a retrospective synthesis, not a new pre-registered confirmatory study. The machine-readable research ledger records source, split, checkpoint and training-configuration provenance. Candidate availability does not establish checkpoint usage: the ten main five-architecture checkpoints are separately bound to inference code and SHA-256 hashes. Recorded manifests, source lines and unsuccessful analyses are preserved. The only recovered paper Git commit includes both historical architecture-family protocols and results; it does not establish a pre-result commit sequence. Filesystem timestamps and protocol assertions cannot close that gap.

The original historical architecture-family ranking-primary hypothesis preceded the later narrative emphasis on local observability according to the documents, but a complete independently verifiable chronology is unavailable. The reversal-control amendment followed an unfavorable one-case smoke outcome. Its construct-validity rationale is recorded; it is not retroactively called outcome-independent. external MRA bright-anchor support was motivated after inspecting TopCoW results; CAS was a subsequent fixed-rule transfer to another public challenge training collection, not its hidden test set. All analyses now using these cohorts are aware of their historical results.

The fastMRI split contains 112/24/24 training/validation/test series and 111/24/24 corresponding patient IDs. One patient overlaps training and validation; test IDs are disjoint. IXI has 404/58/115 disjoint IDs. Eight NingXia patient folds have no within-fold train/validation/test patient overlap. These checks verify local identifiers, not cross-dataset patient linkage.

A corrected influence definition. Earlier manuscript text described averaging response cosines. Both the main HAT/SwinIR implementation and the historical architecture-family new-architecture implementation actually sum shared-anchor response inner products and normalize by *all*-anchor response energy, as in the revised main equation. Historical values remain unchanged. For example, responses $(1, 0)$ on a shared anchor and $(0, 3)$ versus $(0, 4)$ on two different unshared anchors give embedded cosine $1/\sqrt{170}$, whereas the shared-only response cosine is one. Omitting the unshared response energy would define a materially different statistic.

The historical original clean HAT/SwinIR correlation pools plane/model observations against a fixed endpoint (phase-0 MAE for fastMRI/NingXia; disjoint-pair mean MAE for IXI) while resampling examination/patient clusters. It must not be described as an examination-level point correlation. The five-architecture ranking analysis instead first aggregates within examinations. Those estimands remain separate in the revised figures.

Reference self-pair sensitivity. Final source review also found that the historical broad-grid summaries include the reference-versus-itself context, for which overlap count is two, influence is one, and error cosine is one. We retain those original tables below but add an explicitly post-hoc integrity sensitivity excluding all 966 such rows. The remaining 7,728 rows yield influence-minus-count alignment advantages .061/.085 on fastMRI, .128/.118 on IXI, and .070/.060 on NingXia (HAT/SwinIR). All six-stratum simultaneous intervals are positive. The bootstrap uses 5,000 examination/patient draws, shared across models within a cohort and independent streams across cohorts. Weighted-rank calculations are verified against direct repeated-row resampling. The preserved historical non-self figure and table below use this result; the final main Figure 3 and Table 2 instead report the new acquired-only experiment. Historical fixed-one-overlap estimates are unchanged. This strengthens a scoped beyond-count association, not a prospective or causal claim.

Claim hierarchy. Level A is replicated local error-observability loss in the evaluated designs. Level B comprises influence beyond count where stable, case-ranking effects, and the matched-geometry interpretation. Level C comprises universal scalar superiority, prospective acquisition benefit, universal target-free neural mechanism validation, and non-MRI generalization. Level A denotes reproducibility in these data, not prespecification or clinical validation. Exact claim-to-experiment, unit, inference and figure mappings are supplied in the final reproducibility and evidence map.

1.3. Scope and Evidence Contract

This supplement reports protocol details, full strata, and every result that materially limits a submitted claim. Experiments are included according to relevance, not favorability. The manuscript does not claim a new reconstruction architecture, calibrated uncertainty, universal estimator superiority, plane-invariant influence, local failure localization, or prospective acquisition benefit.

The independent units are 24 fastMRI examinations, 115 IXI examinations, and eight NingXia patients containing 22 series. All NingXia intervals resample patients with subordinate series, planes, models, and contexts retained. Missing target values are not subtracted in a disagreement score, but its evaluation support can be target-derived. The historical architecture studies use a ground-truth 1% mask; the recovered bank uses the target model's prediction mask, not ground truth. Anchor-CV uses held-out acquired anchors (including their masks); adjacent MAE uses masks from acquired pairs. These information regimes must not be conflated.

1.4. Data Attribution and Unresolved Author Documentation

The brain T2 collection is locally catalogued as NYU fastMRI and converted from local DICOM series to NIfTI. The conversion manifest and physical-position/pixel-order checks trace the analyzed files, but do not authenticate an official download release or accession. We retain the fastMRI initiative citation required by its data portal, and add the brain challenge publication for brain-dataset context; we do not claim that our local split is the official challenge split. The author must confirm the precise release, access agreement and image-publication permission before submission. NYU fastMRI investigators supplied the attributed resource, without participating in this analysis or manuscript; this attribution is conditional on confirmation of local provenance.

IXI is attributed to the Biomedical Image Analysis Group at Imperial College London. Its public T2 NIfTI collection is described on the Brain Development website and distributed under CC BY-SA 3.0. Our local training/validation/test partition is separate from the source collection. No unsupported dataset publication year is assigned.

NingXia denotes a locally retained knee T2 collection with eight patient groups and 22 analyzed series. Acquisition institution, data custodian, release/allowed-use agreement, de-identification process, IRB approval or exemption, and consent or consent-waiver status are not established by the repository documents reviewed. Local patient-group splitting establishes analysis clustering, not ethics authorization or de-identification. **Author confirmation required before submission:** supply those records and the verified statement here; no approval number, consent waiver or permission is inferred. Direct identifiers found in raw directory naming are not reproduced in the manuscript or review package, and de-identification of the research extract cannot presently be certified.

TopCoW and CAS remain public-challenge training-collection analyses with locally available labels, not hidden-test evaluations. Dataset citations and the remaining local permission/release checks are listed in the final evidence map. No new subject data were acquired for this submission patch.

2. Theory and Framework

2.1. Framework Diagram and Full Covariance Derivation

The accompanying NumPy implementation, the final reproducibility and evidence map, accepts a frozen estimator callable, physical anchor dictionaries, declared support and scale provenance, perturbation settings, and optional retrospective targets. It implements paired predictions, the all-anchor-energy influence statistic, and separate error evaluation. Model freezing and clustered aggregation remain

caller responsibilities. Tests cover acquired-only target-mutation invariance, inconsistent shared-anchor rejection, energy closure, and a linear coefficient reference including perturbation-sign reversal. Reproduction instructions are in the final reproducibility and evidence map.

Condition on fixed nominal inputs and a fixed target y. Write $e_a = b_a + J_a\xi_a$ and $e_b = b_b + J_b\xi_b$, where the perturbations have zero mean and finite second moments. Let $\Sigma_a = \mathbb{E}\xi_a\xi_a^\top$, $\Sigma_b = \mathbb{E}\xi_b\xi_b^\top$, and $C = \mathbb{E}\xi_a\xi_b^\top$, with the joint block covariance positive semidefinite. Then

$$\mathbb{E}\langle e_a, e_b\rangle = \langle b_a, b_b\rangle + \operatorname{tr}(J_a C J_b^\top), \tag{1}$$

$$\tau = \mathbb{E}T = \tfrac{1}{2}\{\|b_a\|^2 + \|b_b\|^2 + \operatorname{tr}(J_a\Sigma_a J_a^\top) + \operatorname{tr}(J_b\Sigma_b J_b^\top)\}. \tag{2}$$

Changing C from zero to a valid cross-covariance, with biases, Jacobians and marginals fixed, changes the first quantity by $\kappa = \operatorname{tr}(J_a C J_b^\top)$ and leaves τ unchanged. Therefore

$$\Delta\mathbb{E}\|e_a - e_b\|^2 = -2\kappa, \tag{3}$$

$$\Delta\mathbb{E}\|(e_a - e_b)/2\|^2 = -\kappa/2, \tag{4}$$

$$\Delta\mathbb{E}\|(e_a + e_b)/2\|^2 = +\kappa/2. \tag{5}$$

For $\tau > 0$, the ratio $\mathbb{E}\|(e_a - e_b)/2\|^2/\tau$ decreases by $\kappa/(2\tau)$ when $\kappa > 0$. This ratio is not the expected samplewise fraction, and positive κ does not establish a monotone change in each realized error cosine or each examination's rank.

Physical-anchor interpretation. If distinct physical anchors have independent perturbations and shared anchors reuse the same perturbation, C has nonzero blocks only for shared identities. Thus $\kappa = \sum_{j \in c_a \cap c_b} \operatorname{tr}(J_{aj}\Sigma_j J_{bj}^\top)$. Aligned sensitivities make these terms positive; orthogonal sensitivities can make sharing irrelevant, and opposing sensitivities can make it increase disagreement. Shared anchor count alone cannot determine the sign or magnitude. If the two contexts are perturbed by independent noise even on an identical anchor, this covariance mechanism is absent despite identical nominal anatomy. Common deterministic bias remains possible in either setting.

Finite responses. For $r_{cjs} = F_c(x + \epsilon\xi_{js}) - F_c(x)$, local differentiability gives $r_{cjs} = \epsilon J_{cj}\xi_{js} + R_{cjs}$. Under the linear model and matched fields,

$$\mathbb{E}\left[\frac{1}{S\epsilon^2}\sum_{s,j}\langle r_{ajs}, r_{bjs}\rangle\right] = \sum_j \operatorname{tr}(J_{aj}\Sigma_j J_{bj}^\top). \tag{6}$$

The empirical normalized ratio is neither this unnormalized trace estimate nor an unbiased covariance estimator. Centering a Rademacher field also changes its covariance from the exact identity to a mean-removal covariance; the trace formula applies to that actual covariance.

For nonlinear outputs write $u_c = J_c\xi_c$. The discrepancy from the linearized inner product is bounded in expectation by

$$\sqrt{\mathbb{E}\|b_a + u_a\|^2\,\mathbb{E}\|R_b\|^2} + \sqrt{\mathbb{E}\|b_b + u_b\|^2\,\mathbb{E}\|R_a\|^2} + \sqrt{\mathbb{E}\|R_a\|^2\,\mathbb{E}\|R_b\|^2}. \tag{7}$$

The proposition is exact for the assumed linear model and approximate only where these remainder terms are controlled. Passing a rank-stability gate is not such a bound. Clipping and changes in active nonlinear regions can invalidate a global Jacobian interpretation. Actual phase changes need not preserve the proposition's biases or marginals; the experimental contrast remains necessary.

Numerical verification covers aligned, orthogonal and opposing sensitivity matrices and cross-correlations $-.5, 0, .5, 1$, with 100,000 Gaussian draws per setting. All twelve predicted inner products and disagreement energies agree within five Monte Carlo standard errors. These checks verify factors and counterexamples, not new natural-image evidence or a clinical sampling distribution.

3. Protocols and Eligibility

3.1. Complete final-analysis protocols

Cross-architecture sharing compares A_0/B_0 and identical A_0/B_2 for every ordered $A \neq B$ among HAT, SwinIR, I3Net, SAINT-style and CycleINR-style. No TTA is used. fastMRI targets are $\{3, 5, 7, 9, 11\}$; IXI uses the unchanged deterministic residue-balanced geometry. Both conditions use the same target-derived 1% support. Five-plane means form examination observations. Primary ΔO and secondary alignment/disagreement differences average examinations. Fixed-reference and condition-specific paired-error rankings correlate examination means separately. The two cohorts are not pooled.

The acquired-only probe selects three rounded interior quartiles of $[3, \text{depth} - 4]$, with geometry-only deduplication. Its six acquired offsets are $\{-3, -2, -1, +1, +2, +3\}$; the target is absent. A common Q1 offset and Q99–Q1 range use finite nonzero acquired intensities; normalized values are clipped to $[0, 1]$. The mask is the union of the nearest acquired planes' 1% foregrounds. Both mask and normalizer are fixed before perturbations and shared across contexts. The common normalizer itself is shared preprocessing. Empty or invalid inputs would remain excluded with reasons; no case was removed by outcome.

Reference $(-2, +2)$ is compared with $(-2, +1)$, $(-2, +3)$, $(-1, +2)$, $(-3, +2)$, $(-1, +1)$, $(-1, +3)$, $(-3, +1)$, $(-3, +3)$. The self-pair is absent. Four centered Rademacher fields use seeds 6101–6104 on the 256×256 input grid. Primary amplitude is .01; .005/.02/$-$.01 are all retained. FP32 without AMP/TF32, input/output clipping and

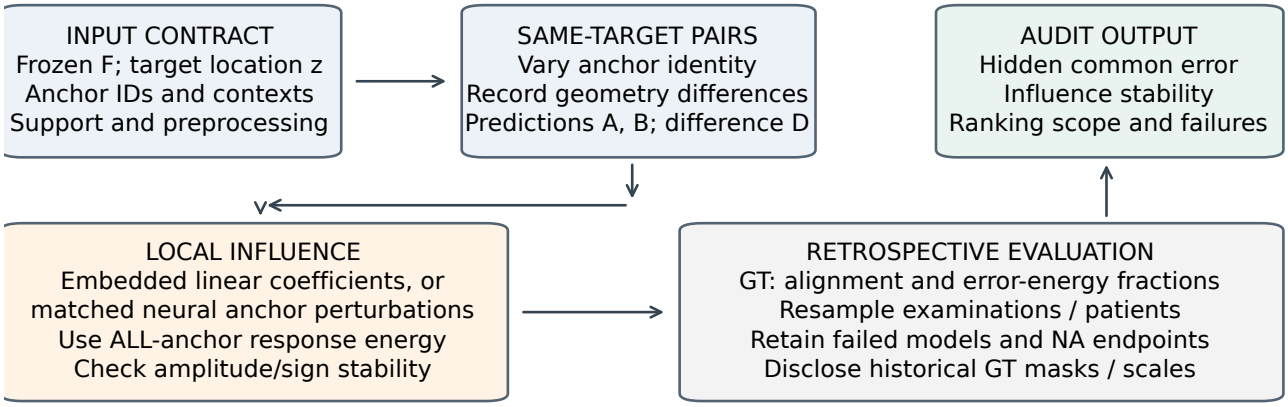


Figure 1. The reusable Measurement-Influence Reliability Audit. The estimator is frozen. Physical anchor identities, support, preprocessing and source of each quantity are part of the input contract. Targets enter retrospective error evaluation; historical GT-mask and full-examination-scale exceptions are disclosed rather than relabeled as deployment inputs. This is a framework diagram, not a proposed reconstruction architecture.

resizing are fixed. Responses and hashes are saved before separate targets are accessed. Error outcomes use the same acquired support. Broad points pool contexts/planes; fixed-count rank residuals adjust span/asymmetry and examination. Examination resampling retains subordinate observations and reranks repeated observations. Weighted computations match direct bootstrap expansion.

Each HAT/SwinIR stratum passes only if all three alternative-versus-primary rank correlations reach the existing .85 threshold; .70 remains the historical catastrophic floor. The original six-stratum aggregate gate is NA without an acquired-only NingXia replication. The .90 threshold belongs to a separate historical new-architecture protocol and is not substituted here.

For the perturbation experiment, sort IXI IDs by SHA-256 of `MIRA-final-proposition:` plus ID and take the first twelve; use the depth-only central target, both models and all four exactly-one-anchor contexts. Perturb only the common anchor, holding other inputs, masks and scales fixed. Four fields and their negatives define eight equal-weight samples. Coupled outputs pair equal field indices; independent outputs average the entire Cartesian product, giving exactly equal empirical marginals. All amplitudes .0025/.005/.01/.02 remain. The odd response at .0025 supplies a fixed linear prediction scaled by squared amplitude. Relative error divides by the magnitude of that prediction. Reference response cosine $\geq .1$, $\leq -.1$ and intermediate values label aligned, opposing and near-orthogonal examples descriptively, without eligibility selection.

All final intervals use 5,000 paired examination draws. Seeds are 18001/18002 for cross-architecture fastMRI/IXI; 18003/18004 for acquired-only IXI/fastMRI; 18005 for perturbation validation. Maximum standardized bootstrap deviation yields simultaneous intervals over 20 ordered pairs per cohort/endpoint, two models per acquired-only cohort/endpoint, or two models times four amplitudes. Pointwise intervals and finite-draw counts remain in the CSVs. These families do not correct the full retrospective search history.

3.2. Objective Qualitative Selection and Precision Check

The main example is selected from the historical complete fastMRI SwinIR table. Rank examinations by their mean disjoint-minus-shared observable gain, tie by ID, and take the lower median (12th of 24). Within that case select the target whose gain is nearest its case mean, tie by z. This produces case 103425595595, $z = 9$. A separate seed-20260911 random draw from the other examinations produces case 101089005814, $z = 7$. The full ranking and selection file were saved before CPU inference. Neither visual appearance nor a newly optimized stable-wrong threshold selects the examples.

Both cases use the hashed historical SwinIR checkpoint with direct target inference using the original wrapper's context normalization, resizing and residual prediction. CPU float32 reference outputs differ from stored GPU mixed-precision outputs by MAE .118/.153 and maximum absolute difference 2.694/2.442 in native units. Thus no exact reproduction claim is made for these new illustrations. Observable fractions differ by less than 4.6×10^{-5} from the historical table. Cohort statistics remain the original values. Original arrays, masks, common display ranges and selection hashes are retained.

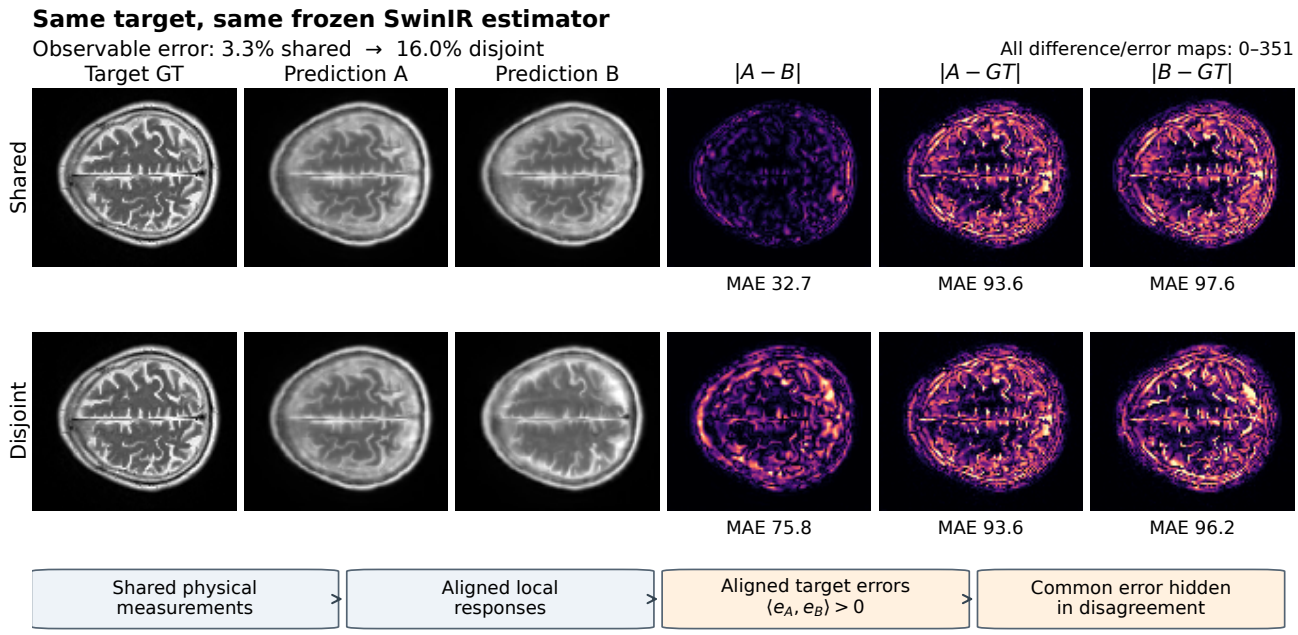


Figure 2. Seeded random supplemental example under the same rendering rule as the main figure. All difference/error maps share a scale; there is no manually selected crop. The reference prediction and target are identical across rows. Shared and disjoint observable fractions are 3.34% and 16.00%. The schematic chain is conditional on aligned sensitivities and does not assert that every shared pair must fail.

3.3. Definitions and Documented Protocol

For predictions $f_a = y + e_a$ and $f_b = y + e_b$, shared and observable error fractions are

$$S = \frac{\|\bar{e}\|^2}{\|\bar{e}\|^2 + \|\delta\|^2}, \qquad O = \frac{\|\delta\|^2}{\|\bar{e}\|^2 + \|\delta\|^2}, \tag{8}$$

where $\bar{e} = (e_a + e_b)/2$ and $\delta = (e_a - e_b)/2$. Numerical identity error is below 4.42×10^{-10} in the stored audit.

For nonlinear models, four frozen seeds (6101–6104) generate centered Rademacher fields at 1% of the globally normalized case range. Each shared anchor is perturbed separately. The influence numerator sums identical-field response inner products over shared anchors and seeds, while its denominator uses the geometric mean of all-anchor response energies on the two sides, including unshared anchors. The broad grid uses reference offsets $(-2, +2)$ and eight target-preserving comparators. Span and asymmetry are explicit covariates; the complete offset list is stored with the experiment registry. The fixed-count subset contains the four comparators with exactly one common anchor.

3.4. Historical Architecture-Family Protocol

The architecture hypothesis and falsification conditions are recorded in local documents; their pre-outcome timing is not independently established. Phase-0 anchors are $\{0, 4, 8, 12\}$, phase-2 anchors are $\{2, 6, 10, 14\}$, and common missing targets are $\{3, 5, 7, 9, 11\}$. The clean shared control applies an in-plane horizontal flip to phase-0 inputs and maps outputs back. A one-case smoke using through-plane reversal was rejected after its one-case outcome exposed order non-equivariance; documents report that this preceded full-cohort inference; its outputs are retained but excluded. Regenerated phase-0 outputs match stored predictions exactly for every one of 24 cases and all five architectures (pixel MAE and maximum absolute difference zero).

I3Net uses four regular anchors and outputs 13 planes. SAINT consumes a left/right pair plus relative and absolute position maps. The internal CycleINR-style model linearly densifies four regular anchors and queries a coordinate-conditioned residual network. The adapted SAINT and CycleINR-style implementations are not claimed as exact official reproductions. All interventions occur at physical anchor identity; no architecture is distorted into another input interface.

The following historical table preserves masked/common-support recovery, including its missing cells. Those missing cells do not mean the native baseline runs are absent: the complete native 13-model, three-cohort comparison and all 111 configurations are retained elsewhere in this supplement. The two supports are not interchangeable and no method is labeled ours.

4. Complete Primary and Supporting Results

4.1. Complete final primary results

Tables below retain every ordered pair and every registered endpoint. Shared/disjoint ranking points use examination means and fixed A-error or condition-specific paired error as explicitly labeled. No null or negative pair is dropped. The acquired-only tables retain all four strata, and the fixed-context table retains every amplitude. Case/context-level values, including all individual approximation failures, are in the corresponding CSVs.

4.2. Interpreting local neural validation

The maximum matched-covariance closure residual is 5.60×10^{-12}, which checks an identity. The independent substantive comparison uses the fixed .0025 odd-response approximation at every amplitude. Median relative error is 1.9–10.6%, but the maximum is 126.4% and the maximum absolute discrepancy is 112.14 native squared-intensity units. There are 86 aligned and ten near-orthogonal fixed model/case/context examples, and no opposing example. All 384 amplitude-specific observations remain. The near-orthogonal examples still have positive covariance; no neural opposing-sign validation is claimed. No post-result amplitude threshold defines success.

4.3. Preserved historical headline figures

These figures retain the original information contracts and estimands. They are not replaced numerically by the new acquired-only or cross-architecture results.

4.4. A Practical Two-Member Ensemble Sharing Contrast

We additionally evaluate fixed linear and four-point Lagrange cubic interpolators on IXI. Pairs L_0–C_0 versus L_0–C_2, and C_0–L_0 versus C_0–L_2, preserve target and estimator composition while varying the companion phase. The score support comes from phase-0 acquired anchors. Complete four-point support is required in both phases, with residue-1/3 equal weighting within cases. In 16-slice fastMRI, only $z = 7$ meets both support requirements, not both residue classes: all 24 cases are excluded by that geometric rule. The initial eligibility assertion and correction of the protocol's mistaken expected target count are logged before proxy summaries; no boundary anchor is fabricated.

Population two-member variance is $|A - B|^2/4$ per pixel, pooled on the common support. It and MAE disagreement are related constructions, not independent experiments. Against fixed reference-model error, variance ranking improves by .0313/.0361, and MAE-disagreement ranking by .0428/.0473. Two-direction simultaneous intervals are positive in this IXI analytic experiment. Paired-error endpoints have the same direction and are separately tabulated. The five-architecture shared control is the complementary deterministic two-view TTA construction. Neither result establishes failure or success of all trained ensembles, stochastic posterior samplers, or calibrated uncertainty systems.

4.5. Material Moved from the Main Paper

The full native baseline table and retrospective practical-consequence plot move here to keep the main paper centered on phenomenon, influence and falsification. All 13 methods and 111 configurations remain, as do 20 training-budget comparisons and every adverse result. The historical common-support table below is a different evaluation contract from the complete native master table.

4.6. Exploratory Full-Model-Bank Audit

After inspecting the recovered benchmark, we added an explicitly exploratory breadth audit; it is not part of the documented architecture-family test. On common case support within each cohort and factor, it covers 6–13 models, 483 undirected error pairs, and 807 directed target–member disagreement pairs. Independent bootstrap units are examinations for fastMRI/IXI and patient folds for NingXia.

Figure 7 shows three results. First, median pairwise case-error rank correlation is .938/.963/.953 for fastMRI $\times 2/\times 4/\times 6$, .991 for IXI $\times 4$, and .992/.980/.972 for NingXia $\times 2/\times 4/\times 6$. All seven clustered 95% intervals have lower bounds above .83. Second, disagreement generally ranks target error, but target–member error dependence does not by itself determine that reliability. Third, a matched cross-family partner is better for several implicit targets yet neutral or worse for several transformer and attention-CNN targets; architecture family is not a universal selection rule.

The NingXia $\times 2$ low tail is important rather than hidden: all 12 pairs with error-rank $\rho < .9$ contain the degraded Res-SRDiff run, whose MAE is 479.32 in the benchmark table. This outlier shows that a failed estimator may produce different errors without producing a generally useful reliability partner. We therefore report the full distribution and do not reinterpret independence as quality.

The equal-weight four-family observable gain is .155 [.148,.162]. Leave-one-family-out estimates are .180 without I3Net, .131 without INR, .161 without SAINT, and .147 without transformers; all intervals exclude zero. The frozen shared-condition thresholds identify stable-wrong rates of 0 for HAT, SwinIR, I3Net, and SAINT and 4/120 (3.33%) for CycleINR; CycleINR has 0/120 under disjoint contexts. This negative prevalence result is not replaced by the earlier earlier stable-wrong definition.

4.7. Distinct-Cohort IXI Five-Architecture Replication

The IXI architecture replication applies the same phase-0/phase-2 clean contrast to all 115 historically reused IXI

Table 1. fastMRI: observable fraction. All ordered pairs; simultaneous family size 20.

Reference A	Companion B	Shared	Disjoint	Disjoint minus shared [simultaneous 95% CI]
HAT	SWINIR	0.048	0.208	0.160 [0.146,0.174]
HAT	I3NET	0.131	0.229	0.098 [0.088,0.108]
HAT	SAINT	0.079	0.209	0.130 [0.116,0.143]
HAT	CYCLEINR	0.165	0.281	0.116 [0.103,0.128]
SWINIR	HAT	0.048	0.205	0.157 [0.143,0.170]
SWINIR	I3NET	0.148	0.234	0.086 [0.075,0.098]
SWINIR	SAINT	0.100	0.209	0.109 [0.093,0.125]
SWINIR	CYCLEINR	0.139	0.250	0.111 [0.094,0.127]
I3NET	HAT	0.131	0.220	0.089 [0.076,0.101]
I3NET	SWINIR	0.148	0.225	0.077 [0.064,0.091]
I3NET	SAINT	0.101	0.208	0.107 [0.097,0.117]
I3NET	CYCLEINR	0.238	0.301	0.062 [0.052,0.072]
SAINT	HAT	0.079	0.206	0.127 [0.112,0.141]
SAINT	SWINIR	0.100	0.209	0.108 [0.092,0.125]
SAINT	I3NET	0.101	0.212	0.111 [0.101,0.122]
SAINT	CYCLEINR	0.208	0.285	0.078 [0.065,0.090]
CYCLEINR	HAT	0.165	0.277	0.112 [0.095,0.129]
CYCLEINR	SWINIR	0.139	0.254	0.115 [0.094,0.135]
CYCLEINR	I3NET	0.238	0.306	0.068 [0.054,0.082]
CYCLEINR	SAINT	0.208	0.283	0.076 [0.057,0.094]

Table 2. fastMRI: error alignment. All ordered pairs; simultaneous family size 20.

Reference A	Companion B	Shared	Disjoint	Disjoint minus shared [simultaneous 95% CI]
HAT	SWINIR	0.906	0.586	-0.320 [-0.348,-0.292]
HAT	I3NET	0.746	0.552	-0.195 [-0.216,-0.174]
HAT	SAINT	0.845	0.587	-0.258 [-0.286,-0.231]
HAT	CYCLEINR	0.702	0.462	-0.240 [-0.265,-0.216]
SWINIR	HAT	0.906	0.594	-0.312 [-0.339,-0.285]
SWINIR	I3NET	0.720	0.547	-0.174 [-0.198,-0.149]
SWINIR	SAINT	0.808	0.591	-0.217 [-0.249,-0.185]
SWINIR	CYCLEINR	0.746	0.520	-0.226 [-0.258,-0.193]
I3NET	HAT	0.746	0.567	-0.180 [-0.205,-0.154]
I3NET	SWINIR	0.720	0.561	-0.159 [-0.186,-0.132]
I3NET	SAINT	0.802	0.588	-0.214 [-0.233,-0.194]
I3NET	CYCLEINR	0.575	0.439	-0.136 [-0.156,-0.115]
SAINT	HAT	0.845	0.591	-0.254 [-0.282,-0.226]
SAINT	SWINIR	0.808	0.588	-0.220 [-0.252,-0.188]
SAINT	I3NET	0.802	0.580	-0.222 [-0.243,-0.201]
SAINT	CYCLEINR	0.628	0.463	-0.165 [-0.191,-0.139]
CYCLEINR	HAT	0.702	0.471	-0.231 [-0.267,-0.195]
CYCLEINR	SWINIR	0.746	0.514	-0.232 [-0.273,-0.190]
CYCLEINR	I3NET	0.575	0.431	-0.144 [-0.172,-0.116]
CYCLEINR	SAINT	0.628	0.469	-0.159 [-0.197,-0.122]

reconstruction-test examinations, with five deterministic targets per examination and 5,000 examination-bootstrap replicates. The historical gate checked one IXI examination per wrapper. The final array audit additionally reproduces all 115 examinations per model with zero maximum absolute difference under the legacy mapping; new SAINT physical coordinates remain a separately declared correction. All five models, examinations, five targets and both conditions remain, without duplicate keys or non-finite metric values.

Every historical pointwise clean and reflection-balanced gain interval is positive, and every model has a positive effect in all 115 examinations. The clean equal-weight family estimate is .286; leave-one-family-out estimates are .281–.297. The signed mirror-difference interval includes zero for

Table 3. fastMRI: disagreement MAE. All ordered pairs; simultaneous family size 20.

Reference A	Companion B	Shared	Disjoint	Disjoint minus shared [simultaneous 95% CI]
HAT	SWINIR	32.561	68.966	36.405 [30.512,42.297]
HAT	I3NET	49.666	64.623	14.957 [12.590,17.323]
HAT	SAINT	39.651	63.827	24.176 [19.954,28.398]
HAT	CYCLEINR	69.079	97.251	28.172 [24.182,32.162]
SWINIR	HAT	32.561	68.394	35.833 [30.035,41.630]
SWINIR	I3NET	54.252	67.907	13.655 [11.270,16.040]
SWINIR	SAINT	45.376	66.128	20.751 [16.576,24.926]
SWINIR	CYCLEINR	62.572	93.295	30.723 [25.738,35.707]
I3NET	HAT	49.666	64.061	14.394 [11.762,17.027]
I3NET	SWINIR	54.252	67.148	12.896 [10.376,15.416]
I3NET	SAINT	41.630	58.546	16.915 [14.279,19.552]
I3NET	CYCLEINR	82.091	94.770	12.678 [10.256,15.101]
SAINT	HAT	39.651	63.958	24.306 [20.003,28.610]
SAINT	SWINIR	45.376	66.666	21.290 [17.164,25.415]
SAINT	I3NET	41.630	58.688	17.057 [14.337,19.778]
SAINT	CYCLEINR	77.427	94.594	17.167 [14.113,20.221]
CYCLEINR	HAT	69.079	95.043	25.964 [21.814,30.114]
CYCLEINR	SWINIR	62.572	92.557	29.984 [24.559,35.409]
CYCLEINR	I3NET	82.091	93.957	11.866 [9.357,14.375]
CYCLEINR	SAINT	77.427	92.818	15.391 [12.023,18.758]

Table 4. fastMRI: fixed-reference examination ranking. All ordered pairs; simultaneous family size 20.

Reference A	Companion B	Shared	Disjoint	Disjoint minus shared [simultaneous 95% CI]
HAT	SWINIR	0.930	0.963	0.033 [-0.157,0.223]
HAT	I3NET	0.954	0.968	0.014 [-0.100,0.128]
HAT	SAINT	0.963	0.965	0.002 [-0.131,0.134]
HAT	CYCLEINR	0.810	0.881	0.070 [-0.118,0.259]
SWINIR	HAT	0.957	0.974	0.017 [-0.135,0.168]
SWINIR	I3NET	0.971	0.980	0.009 [-0.090,0.108]
SWINIR	SAINT	0.962	0.964	0.003 [-0.163,0.168]
SWINIR	CYCLEINR	0.770	0.908	0.137 [-0.149,0.424]
I3NET	HAT	0.919	0.953	0.034 [-0.110,0.178]
I3NET	SWINIR	0.911	0.945	0.034 [-0.139,0.207]
I3NET	SAINT	0.957	0.974	0.017 [-0.090,0.125]
I3NET	CYCLEINR	0.710	0.821	0.111 [-0.117,0.340]
SAINT	HAT	0.955	0.968	0.013 [-0.101,0.127]
SAINT	SWINIR	0.931	0.968	0.037 [-0.136,0.209]
SAINT	I3NET	0.968	0.965	-0.003 [-0.116,0.111]
SAINT	CYCLEINR	0.830	0.858	0.028 [-0.156,0.211]
CYCLEINR	HAT	0.965	0.983	0.018 [-0.110,0.146]
CYCLEINR	SWINIR	0.938	0.976	0.037 [-0.139,0.214]
CYCLEINR	I3NET	0.965	0.990	0.025 [-0.073,0.123]
CYCLEINR	SAINT	0.980	0.984	0.004 [-0.097,0.105]

every model, so the frozen orientation-warning list is empty. This supports physical anchor identity after matching the nearest-distance multiset $\{1, 3\}$ up to reflection. It does not establish exact signed-geometry identity, because the two residue classes are reflection-equivalent rather than identical.

4.8. Controlled Dependence Results

The original pooled-plane ranking and stable-wrong summaries are retained here without changing their endpoints, thresholds or weights. They differ from the uniform local-error estimand used by the revised main Figure 2 and Table 1.

Table 5. fastMRI: condition-specific paired-error examination ranking. All ordered pairs; simultaneous family size 20.

Reference A	Companion B	Shared	Disjoint	Disjoint minus shared [simultaneous 95% CI]
HAT	SWINIR	0.943	0.964	0.021 [-0.160,0.201]
HAT	I3NET	0.931	0.972	0.041 [-0.094,0.175]
HAT	SAINT	0.957	0.965	0.008 [-0.100,0.115]
HAT	CYCLEINR	0.934	0.973	0.039 [-0.081,0.159]
SWINIR	HAT	0.943	0.972	0.029 [-0.140,0.197]
SWINIR	I3NET	0.932	0.961	0.029 [-0.120,0.178]
SWINIR	SAINT	0.947	0.968	0.021 [-0.147,0.189]
SWINIR	CYCLEINR	0.886	0.977	0.090 [-0.126,0.307]
I3NET	HAT	0.931	0.974	0.043 [-0.098,0.183]
I3NET	SWINIR	0.932	0.973	0.041 [-0.130,0.212]
I3NET	SAINT	0.964	0.982	0.017 [-0.104,0.139]
I3NET	CYCLEINR	0.910	0.957	0.047 [-0.099,0.193]
SAINT	HAT	0.957	0.966	0.009 [-0.111,0.128]
SAINT	SWINIR	0.947	0.971	0.024 [-0.130,0.178]
SAINT	I3NET	0.964	0.984	0.020 [-0.093,0.133]
SAINT	CYCLEINR	0.948	0.972	0.024 [-0.099,0.147]
CYCLEINR	HAT	0.934	0.983	0.050 [-0.076,0.175]
CYCLEINR	SWINIR	0.886	0.988	0.102 [-0.089,0.293]
CYCLEINR	I3NET	0.910	0.986	0.076 [-0.082,0.233]
CYCLEINR	SAINT	0.948	0.982	0.034 [-0.102,0.170]

Table 6. IXI: observable fraction. All ordered pairs; simultaneous family size 20.

Reference A	Companion B	Shared	Disjoint	Disjoint minus shared [simultaneous 95% CI]
HAT	SWINIR	0.023	0.309	0.285 [0.278,0.292]
HAT	I3NET	0.144	0.328	0.183 [0.173,0.193]
HAT	SAINT	0.036	0.312	0.275 [0.268,0.283]
HAT	CYCLEINR	0.125	0.337	0.212 [0.206,0.218]
SWINIR	HAT	0.023	0.308	0.285 [0.278,0.292]
SWINIR	I3NET	0.149	0.331	0.183 [0.173,0.193]
SWINIR	SAINT	0.035	0.316	0.281 [0.274,0.289]
SWINIR	CYCLEINR	0.113	0.337	0.224 [0.218,0.230]
I3NET	HAT	0.144	0.324	0.180 [0.169,0.191]
I3NET	SWINIR	0.149	0.328	0.180 [0.169,0.191]
I3NET	SAINT	0.149	0.329	0.180 [0.168,0.191]
I3NET	CYCLEINR	0.237	0.368	0.131 [0.121,0.141]
SAINT	HAT	0.036	0.311	0.275 [0.268,0.282]
SAINT	SWINIR	0.035	0.316	0.281 [0.274,0.288]
SAINT	I3NET	0.149	0.332	0.182 [0.172,0.193]
SAINT	CYCLEINR	0.107	0.333	0.226 [0.220,0.232]
CYCLEINR	HAT	0.125	0.336	0.211 [0.206,0.217]
CYCLEINR	SWINIR	0.113	0.336	0.223 [0.218,0.229]
CYCLEINR	I3NET	0.237	0.368	0.132 [0.122,0.141]
CYCLEINR	SAINT	0.107	0.332	0.226 [0.220,0.231]

The original order-reversal control is not used as the cleaner dependence contrast. It supports the complete covariance chain on IXI and NingXia, but several cosine/shared/observable arrows reverse on fastMRI because order reversal changes more than anchor dependence. The horizontal-flip clean control preserves anchor identities, target, distances, count, z order, mask, and preprocessing and is the cleaner historical TTA intervention. The final cross-architecture primary experiment uses no TTA.

Metric robustness was assessed over MAE, RMSE/MSE, SSIM error, gradient MAE, low-pass MAE, and residual high-frequency MAE for three physical planes, three cohorts, and two architectures. All 42 disjoint-minus-shared point differences are positive. All seven IXI intervals exclude

Table 7. IXI: error alignment. All ordered pairs; simultaneous family size 20.

Reference A	Companion B	Shared	Disjoint	Disjoint minus shared [simultaneous 95% CI]
HAT	SWINIR	0.953	0.392	-0.562 [-0.577,-0.547]
HAT	I3NET	0.751	0.376	-0.375 [-0.394,-0.356]
HAT	SAINT	0.928	0.385	-0.543 [-0.558,-0.528]
HAT	CYCLEINR	0.772	0.340	-0.432 [-0.445,-0.419]
SWINIR	HAT	0.953	0.393	-0.561 [-0.576,-0.545]
SWINIR	I3NET	0.744	0.369	-0.375 [-0.393,-0.356]
SWINIR	SAINT	0.931	0.377	-0.554 [-0.569,-0.539]
SWINIR	CYCLEINR	0.795	0.339	-0.456 [-0.470,-0.443]
I3NET	HAT	0.751	0.374	-0.377 [-0.398,-0.356]
I3NET	SWINIR	0.744	0.367	-0.377 [-0.398,-0.356]
I3NET	SAINT	0.744	0.366	-0.377 [-0.399,-0.356]
I3NET	CYCLEINR	0.598	0.302	-0.296 [-0.317,-0.276]
SAINT	HAT	0.928	0.387	-0.541 [-0.556,-0.526]
SAINT	SWINIR	0.931	0.378	-0.553 [-0.568,-0.538]
SAINT	I3NET	0.744	0.369	-0.375 [-0.394,-0.356]
SAINT	CYCLEINR	0.807	0.348	-0.459 [-0.473,-0.446]
CYCLEINR	HAT	0.772	0.348	-0.424 [-0.436,-0.412]
CYCLEINR	SWINIR	0.795	0.347	-0.448 [-0.460,-0.437]
CYCLEINR	I3NET	0.598	0.310	-0.288 [-0.305,-0.271]
CYCLEINR	SAINT	0.807	0.355	-0.452 [-0.464,-0.440]

Table 8. IXI: disagreement MAE. All ordered pairs; simultaneous family size 20.

Reference A	Companion B	Shared	Disjoint	Disjoint minus shared [simultaneous 95% CI]
HAT	SWINIR	12.802	47.636	34.834 [29.981,39.688]
HAT	I3NET	25.363	42.163	16.799 [14.023,19.575]
HAT	SAINT	16.663	48.019	31.356 [26.933,35.778]
HAT	CYCLEINR	32.883	58.337	25.454 [21.767,29.141]
SWINIR	HAT	12.802	47.512	34.710 [29.891,39.529]
SWINIR	I3NET	26.156	42.728	16.572 [13.834,19.310]
SWINIR	SAINT	16.435	48.484	32.049 [27.541,36.557]
SWINIR	CYCLEINR	30.953	58.390	27.438 [23.541,31.335]
I3NET	HAT	25.363	41.824	16.461 [13.620,19.302]
I3NET	SWINIR	26.156	42.482	16.325 [13.511,19.140]
I3NET	SAINT	27.134	42.754	15.621 [12.861,18.381]
I3NET	CYCLEINR	42.289	54.835	12.546 [10.251,14.841]
SAINT	HAT	16.663	47.983	31.320 [26.922,35.718]
SAINT	SWINIR	16.435	48.541	32.106 [27.598,36.614]
SAINT	I3NET	27.134	43.073	15.939 [13.263,18.616]
SAINT	CYCLEINR	30.820	58.205	27.386 [23.449,31.323]
CYCLEINR	HAT	32.883	58.318	25.434 [21.767,29.101]
CYCLEINR	SWINIR	30.953	58.441	27.489 [23.586,31.392]
CYCLEINR	I3NET	42.289	54.936	12.647 [10.444,14.851]
CYCLEINR	SAINT	30.820	58.206	27.386 [23.455,31.317]

zero. SSIM error has weak or negative absolute ranking on fastMRI/NingXia and several smaller-cohort gradient/high-frequency intervals cross zero. The claim is broad error observability, not metric independence.

4.9. Exact Count Falsification and Analytical Influence

The compact analytical ladder uses all valid sign configurations at fixed absolute distances $\{1, 2, 3, 4\}$. Across fastMRI, IXI, and NingXia, overlap count has overall $+.9$ association with error cosine/shared fraction and $-.9$ to -1.0 with ob-

Table 9. IXI: fixed-reference examination ranking. All ordered pairs; simultaneous family size 20.

Reference A	Companion B	Shared	Disjoint	Disjoint minus shared [simultaneous 95% CI]
HAT	SWINIR	0.964	0.992	0.028 [0.001,0.056]
HAT	I3NET	0.969	0.993	0.025 [-0.003,0.053]
HAT	SAINT	0.958	0.991	0.033 [0.002,0.064]
HAT	CYCLEINR	0.938	0.983	0.045 [0.011,0.078]
SWINIR	HAT	0.965	0.993	0.028 [0.001,0.055]
SWINIR	I3NET	0.970	0.994	0.024 [-0.003,0.051]
SWINIR	SAINT	0.961	0.992	0.031 [0.003,0.059]
SWINIR	CYCLEINR	0.939	0.983	0.044 [0.013,0.076]
I3NET	HAT	0.937	0.987	0.050 [0.002,0.098]
I3NET	SWINIR	0.938	0.986	0.049 [0.002,0.095]
I3NET	SAINT	0.932	0.986	0.053 [0.004,0.103]
I3NET	CYCLEINR	0.912	0.970	0.058 [0.008,0.108]
SAINT	HAT	0.962	0.993	0.031 [0.002,0.061]
SAINT	SWINIR	0.964	0.993	0.030 [0.003,0.056]
SAINT	I3NET	0.967	0.994	0.027 [-0.002,0.057]
SAINT	CYCLEINR	0.946	0.985	0.040 [0.010,0.069]
CYCLEINR	HAT	0.964	0.993	0.029 [0.004,0.054]
CYCLEINR	SWINIR	0.964	0.993	0.030 [0.005,0.054]
CYCLEINR	I3NET	0.968	0.994	0.025 [-0.001,0.052]
CYCLEINR	SAINT	0.967	0.993	0.026 [0.003,0.049]

Table 10. IXI: condition-specific paired-error examination ranking. All ordered pairs; simultaneous family size 20.

Reference A	Companion B	Shared	Disjoint	Disjoint minus shared [simultaneous 95% CI]
HAT	SWINIR	0.964	0.994	0.030 [0.004,0.057]
HAT	I3NET	0.958	0.993	0.036 [0.002,0.069]
HAT	SAINT	0.960	0.994	0.034 [0.004,0.064]
HAT	CYCLEINR	0.954	0.993	0.039 [0.009,0.068]
SWINIR	HAT	0.964	0.994	0.030 [0.004,0.057]
SWINIR	I3NET	0.959	0.994	0.035 [0.003,0.067]
SWINIR	SAINT	0.962	0.994	0.032 [0.005,0.059]
SWINIR	CYCLEINR	0.953	0.992	0.039 [0.011,0.067]
I3NET	HAT	0.958	0.995	0.037 [0.003,0.071]
I3NET	SWINIR	0.959	0.995	0.036 [0.004,0.068]
I3NET	SAINT	0.955	0.995	0.040 [0.005,0.075]
I3NET	CYCLEINR	0.954	0.992	0.038 [0.005,0.071]
SAINT	HAT	0.960	0.995	0.034 [0.004,0.064]
SAINT	SWINIR	0.962	0.995	0.032 [0.005,0.060]
SAINT	I3NET	0.955	0.993	0.038 [0.004,0.073]
SAINT	CYCLEINR	0.958	0.993	0.035 [0.008,0.061]
CYCLEINR	HAT	0.954	0.992	0.038 [0.010,0.066]
CYCLEINR	SWINIR	0.953	0.992	0.039 [0.012,0.065]
CYCLEINR	I3NET	0.954	0.990	0.036 [0.005,0.067]
CYCLEINR	SAINT	0.958	0.992	0.033 [0.009,0.058]

servable fraction/disagreement. Every cohort nevertheless shows a zero-to-one reversal. A wider $\{2, 4, 6, 8\}$ ladder reproduces the exception on IXI while NingXia is monotone. This rejects strict count monotonicity without rejecting a broad dependence effect.

For the four main analytical endpoints, clustered intervals for the absolute-correlation improvement of common coefficient mass over count are all above zero, spanning .189–.283. Signed dot product and coefficient cosine agree qualitatively; common mass is reported because it retains unequal shared magnitude.

Table 11. Acquired-only neural influence. Beyond-count and fixed-one-anchor estimates have simultaneous 95% intervals across both models, separately by cohort/endpoint. Stability is the minimum of three amplitude/sign correlations; the historical per-stratum threshold is .85.

Cohort	Model	$\lvert\rho_I\rvert - \lvert\rho_{count}\rvert$ [CI]	Fixed-count partial ρ [CI]	Minimum stability / status
IXI	HAT	0.098 [0.090,0.106]	0.270 [0.194,0.345]	0.930 / PASS
IXI	SwinIR	0.093 [0.084,0.103]	0.265 [0.195,0.334]	0.880 / PASS
fastMRI	HAT	0.103 [0.089,0.118]	0.474 [0.321,0.627]	0.964 / PASS
fastMRI	SwinIR	0.093 [0.080,0.106]	0.365 [0.216,0.514]	0.942 / PASS

Table 12. All acquired-only amplitude/sign stability checks against +.01. The per-stratum threshold is .85; the original six-stratum aggregate gate is NA.

Cohort	Model	Alternative amplitude	Rank stability	Status
fastmri	HAT	0.005	0.9855	PASS
fastmri	HAT	0.02	0.9643	PASS
fastmri	HAT	-0.01	0.9726	PASS
fastmri	SWINIR	0.005	0.9789	PASS
fastmri	SWINIR	0.02	0.9421	PASS
fastmri	SWINIR	-0.01	0.9776	PASS
ixi	HAT	0.005	0.9298	PASS
ixi	HAT	0.02	0.9576	PASS
ixi	HAT	-0.01	0.9348	PASS
ixi	SWINIR	0.005	0.8803	PASS
ixi	SWINIR	0.02	0.9505	PASS
ixi	SWINIR	-0.01	0.9113	PASS

4.10. Full Neural Shared-Influence Results

The corresponding primary shared-fraction and observable-fraction advantages are .040/.040 and .056/.056 on fastMRI, .089/.089 and .081/.081 on IXI, and .049/.049 and .043/.043 on NingXia for HAT/SwinIR. Every interval excludes zero. Direct disagreement improves in all six strata, but both fastMRI intervals cross zero; it is supporting rather than primary evidence.

4.10.1. Fixed-one-overlap test

NingXia does not preserve the expected direction consistently and all intervals cross zero. We therefore do not use wording such as "the direction is preserved but less precise." The actual result is null/mixed under eight-patient fixed-count inference.

4.10.2. Perturbation and plane stability

The perturbation-stability audit varies amplitude (.5/1/2%), positive/negative field, Rademacher/Gaussian field, and 1/2/4/8 directions. All 42 cohort–architecture–condition strata pass the frozen rank-stability threshold; minimum Spearman is .942 and maximum .999. Physical-plane pair stability is .779–.954, and only one of three plane-pair gates passes. This makes the statistic robust to the tested numerical perturbation design but local to a declared physical plane.

4.11. Mechanism Chain

The context-ranking endpoint uses only eight configurations and is not a mediation estimate. Influence-to-ranking is still less stable. The manuscript therefore claims that influence is one determinant of local error alignment/observability, not that it alone determines global case ranking.

4.12. Anchor-CV, Strong Baselines, and Consequence

Anchor-CV exactly equals the masked MAE of half the acquired-slice second difference. Frozen-score reproduction has maximum absolute error 5.68×10^{-14}. Table 25 reports strong baselines that prevent a uniqueness claim.

Partial rank association for Anchor-CV after adjacent MAE, robust scale, and spacing is .333 on fastMRI, $-.059$ on IXI, and $-.100$ on NingXia. Scale normalization reduces/reverses rankings, so the score is not a scale-free invariant. Frozen risk–coverage also has no universal winner: output PC is better on fastMRI/IXI and adjacent MAE is better on NingXia.

The fixed-rule retrospective consequence audit instead matches model identity while allowing factor-specific weights. It pairs common-support $\times 4$ and $\times 2$ MAE for every eligible model on fastMRI and patient-folded NingXia. Anchor-CV is positively associated with benefit for 13/13 fastMRI models and 12/13 NingXia models (median within-

Table 13. Every fixed-context neural amplitude. Effects are in native squared-intensity units; intervals are simultaneous over eight model/amplitude contrasts. Relative approximation error uses the fixed .0025 prediction as denominator.

Model	Amplitude	Observed mean [95% CI]	Fixed-linear mean	Median relative error
HAT	0.0025	-1.803 [-2.835,-0.771]	-1.792	1.90%
HAT	0.005	-7.350 [-11.630,-3.069]	-7.169	5.75%
HAT	0.01	-28.758 [-45.345,-12.172]	-28.676	8.92%
HAT	0.02	-110.005 [-173.455,-46.556]	-114.705	10.61%
SWINIR	0.0025	-1.502 [-2.366,-0.639]	-1.515	2.86%
SWINIR	0.005	-6.028 [-9.453,-2.603]	-6.060	9.65%
SWINIR	0.01	-24.178 [-37.927,-10.428]	-24.240	10.35%
SWINIR	0.02	-94.647 [-148.847,-40.446]	-96.958	9.83%

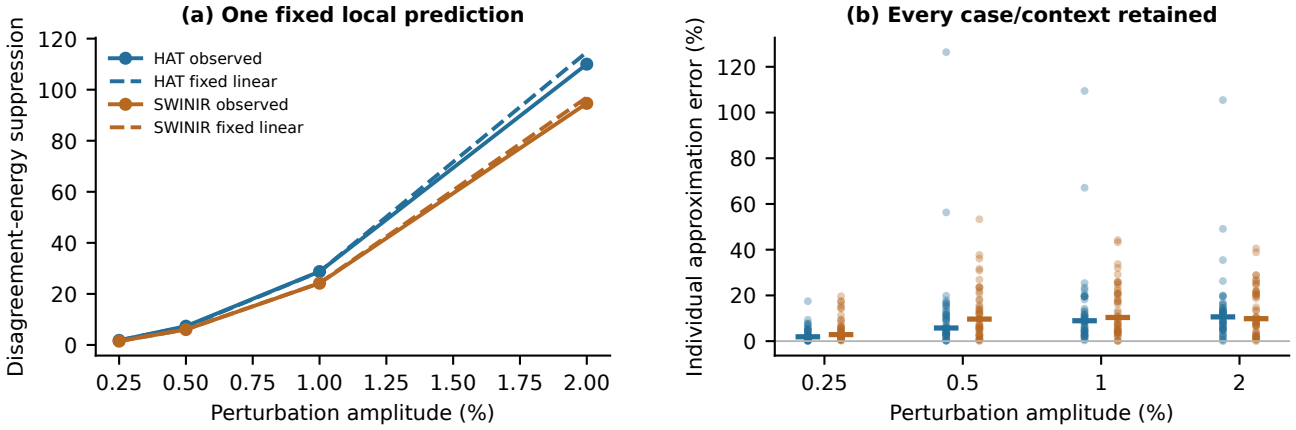


Figure 3. Fixed-context neural covariance experiment. (a) Mean observed coupled-minus-independent disagreement compared with the single .0025 odd-response prediction scaled by squared amplitude. (b) All individual relative approximation errors, including the 126.4% boundary. The exact same-amplitude covariance closure is a separate identity check.

model $\rho = .851/.727$); adjacent MAE gives 13/13 and 12/13 (.802/.748), and candidate-bank disagreement gives 11/11 and 10/11 (.835/.710). The smaller bank denominator is structural: the precomputation includes neural targets only; Linear/Cubic are candidate members but not target rows. The only non-positive model is the severely degraded NingXia Res-SRDiff run; all three intervals cross zero. No model is removed by result direction.

At 10/20/30/50% selection budgets, model-median oracle-normalized gains remain positive for all three scores in both cohorts. Individual NingXia Res-SRDiff policies can fall below random, so allocation is not universally beneficial. These analyses use already reconstructed dense acquisitions and cannot identify the causal scanner-time benefit of changing acquisition density.

Anchor-CV takes .42/3.97/3.93 ms CPU on representative fastMRI/IXI/NingXia examinations with zero neural calls. Output PC is cheap only after predictions exist. Neither Anchor-CV nor PC localizes the worst physical plane; their supported use is examination triage.

4.13. Exploratory Robustness and Information Audit

These analyses were designed after inspecting existing results. They add no independent cohort and are not prospectively registered. All five architectures, both brain cohorts, all eligible benefit models, and adverse outcomes are retained. Bootstrap count is 10,000; the seed is 20260908.

Aggregation and simultaneous inference. We recompute observable gain with all four combinations of within-case and across-case mean/median aggregation. Every model retains a positive five-model simultaneous interval in every combination and cohort. Intervals use the maximum absolute bootstrap deviation divided by each estimate's bootstrap standard deviation, with shared examination draws across models. They are approximate bootstrap simultaneous intervals, not exact small-sample coverage guarantees. Historical fastMRI local gain uses mean/mean aggregation, whereas ranking uses within-case medians; IXI uses medians. Cohorts remain separate.

The more fragile endpoint is ranking. Under historical

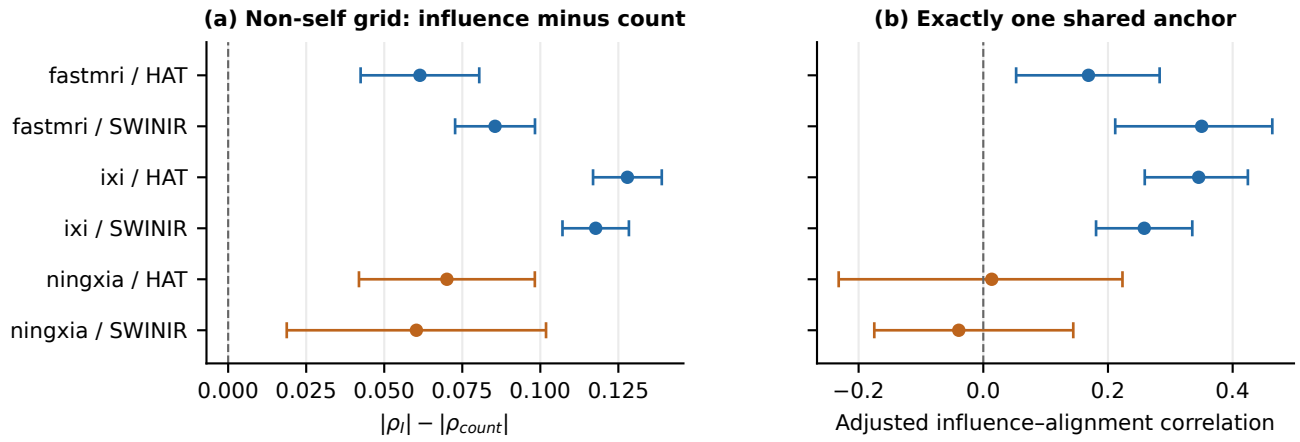


Figure 4. Historical target-informed non-self influence advantage with six-stratum simultaneous intervals, alongside the fixed-count test with historical pointwise intervals. The NingXia null/mixed result remains.

Table 14. Non-self broad grid and fixed-count influence. Broad-grid advantage uses six-stratum simultaneous 95% intervals; fixed-count estimates retain historical pointwise clustered 95% intervals. Both analyses preserve every cohort/model.

Cohort	Model	ρ_{count}	ρ_I	Non-self advantage [CI]	Fixed-one partial ρ [CI]
fastmri	HAT	0.704	0.766	0.061 [0.042,0.080]	0.169 [0.053,0.283]
fastmri	SWINIR	0.699	0.784	0.085 [0.073,0.098]	0.350 [0.212,0.464]
ixi	HAT	0.577	0.705	0.128 [0.117,0.139]	0.346 [0.259,0.424]
ixi	SWINIR	0.573	0.691	0.118 [0.107,0.128]	0.258 [0.181,0.335]
ningxia	HAT	0.620	0.690	0.070 [0.042,0.098]	0.014 [-0.232,0.223]
ningxia	SWINIR	0.609	0.669	0.060 [0.019,0.102]	-0.039 [-0.175,0.144]

within-case median aggregation, simultaneous intervals exclude zero for IXI HAT, SwinIR and SAINT only; I3Net and CycleINR include zero. No fastMRI ranking interval excludes zero under simultaneous inference, including CycleINR. Historical pointwise intervals remain reported; they answer a different multiplicity question. Averaging the two conditions' paired errors to obtain a common ranking endpoint gives the same three-versus-zero simultaneous pattern. This common endpoint remains post-intervention and is not a fixed phase-0 error estimate.

Densification and current-error coupling. For the existing 598 rows, $b = E_4 - E_2$ is checked exactly and models are paired on the same case, fold and support. Matching labels does not hold scale-specific checkpoints fixed. Since b contains E_4, raw score–benefit association need not establish incremental information beyond current difficulty. We regress ranks of score and benefit on the rank of E_4 and correlate residuals, reranking within each bootstrap sample. NingXia resamples eight patients with subordinate series retained; degenerate partial-correlation replicates are recorded rather than silently counted as zero.

Recovered IXI same-architecture pair. The existing I3Net $\times 2$ and $\times 4$ profiles each contain 115 outputs and distinct checkpoints. All 7,334 common odd target planes are re-evaluated with identical physical affines and a common target-derived 1% mask; case MAE averages planes before inference. Mean MAE is 33.65 at $\times 4$ and 25.62 at $\times 2$, with positive benefit for 115/115 examinations. Linear interpolation on this support gives 57.36 and 39.21. These numbers are not interchangeable with the benchmark's all-generated-plane averages.

Anchor-CV and adjacent MAE, computed only from $\times 4$ acquired anchors, give raw benefit correlations .843 [.768,.894] and .845 [.771,.896]. Their partial correlations given current error are .216 [.007,.430] and .227 [.020,.437]. These exploratory pointwise intervals do not imply universal superiority. Dividing the scores by the acquired-anchor 1st–99th percentile range reverses raw-benefit ranking to $-.238$ $[-.404, -.058]$ and $-.236$ $[-.397, -.060]$. At 50% selection normalized Anchor-CV is below random by .920 raw MAE units $[-1.635, -.149]$. The endpoint is deliberately left in raw units: this establishes sensitivity to the scoring scale, not failure of every jointly normalized evaluation.

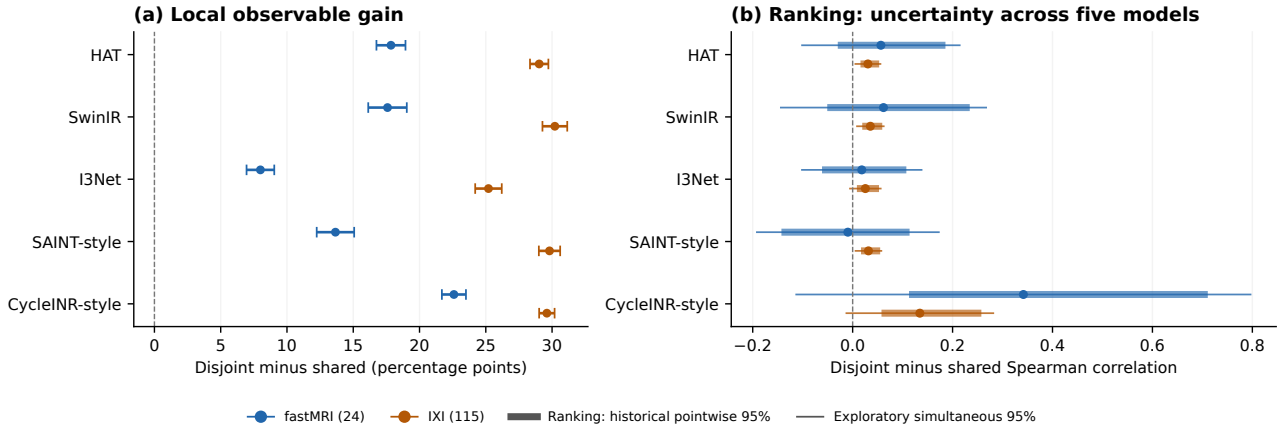


Figure 5. Historical same-model TTA replication: local effects versus examination ranking. Local fastMRI points use means and IXI medians; ranking uses within-examination medians. Five-model simultaneous intervals preserve the weak/null ranking results. Legacy SAINT coordinates are retained in this historical figure.

Table 15. IXI two-member analytical ensemble, 115 examinations. Contrasts are disjoint minus shared, with simultaneous intervals over the two directions for each displayed endpoint. "Fixed uses the unchanged reference predictor; "paired uses both predictions.

Proxy / metric	Error endpoint	L_0 reference [95% CI]	C_0 reference [95% CI]
Observable gain	—	0.3213 [0.3166,0.3260]	0.3212 [0.3165,0.3259]
MAE disagreement	Fixed	0.0428 [0.0188,0.0668]	0.0473 [0.0220,0.0727]
MAE disagreement	Paired	0.0471 [0.0228,0.0714]	0.0470 [0.0228,0.0712]
Population variance	Fixed	0.0313 [0.0122,0.0503]	0.0361 [0.0158,0.0564]
Population variance	Paired	0.0354 [0.0158,0.0550]	0.0356 [0.0159,0.0552]

Algebra and information access. Writing T for mean paired squared error and q for shared fraction, normalized RMSE disagreement equals $2\sqrt{1-q}$. The historical TMI script actually uses MAE disagreement, so its statistic is $2\kappa\sqrt{1-q}$ with $\kappa = \mathrm{MAE}(a-b)/\mathrm{RMSE}(a-b) \leq 1$. The residual-shape factor must not be omitted. The shared-fraction correlation is structurally coupled and does not independently establish a mechanism.

The historical architecture studies output-score mask uses dense targets. The bank instead uses target-model prediction support; its field name does not denote ground truth. Raw Anchor-CV and adjacent MAE use acquired-only masks. The earlier scale-normalized Anchor-CV uses a full-volume robust range, whereas the new IXI normalization explicitly uses acquired anchors. Deployment requires a separately frozen acquisition-only mask and scale convention. Exact wrapper agreement validates implementation consistency for the tested cases, not freedom from target-support bias or general model correctness.

Reproducibility. The new scripts, source hashes, geometry checks, complete model-specific intervals, leave-one-cluster-out tables and patient-equal sensitivity results are retained in the final reproducibility and evidence map. All reported values are regenerated from the formal CSV records; technical smoke outputs are excluded by declared file identity.

4.14. Complete Baseline Ledger and Training-Budget Sensitivity

The experiment master table contains 13 reconstruction baselines. We recover all 39 three-cohort $\times 4$ rows, all 52 fastMRI/NingXia $\times 2/\times 6$ rows, and all 20 recorded longer-budget rows: 111 configurations and 555 metric values. Original metric summaries agree to their reported four-decimal precision. Two overwritten SAINT summaries at fastMRI $\times 2/\times 6$ are recovered from all 48 preserved output volumes. No run is retrained for this audit.

The native evaluator measures every nonacquired plane on the whole image, averages planes within an examination, and then averages examinations. NingXia first averages series within a held-out patient fold, then averages all eight folds. PSNR uses each target plane's range; SSIM is the full-plane implementation. These are reference-based evaluation metrics, not deployable QA scores. Each configuration has matching case identities and generated-plane counts across its models; this count audit does not independently certify all historical preprocessing. The old masked common-support

Table 16. All 13 master-table baselines at ×4, independently recovered from original metrics. PSNR/SSIM are higher-is-better; MAE is lower-is-better. Full-plane evaluation excludes acquired planes. IXI/fastMRI average examinations; NingXia averages eight patient folds. These values do not replace the masked common-support reliability endpoints.

	IXI (115)			fastMRI (24)			NingXia (8 folds)		
Method	PSNR	SSIM	MAE	PSNR	SSIM	MAE	PSNR	SSIM	MAE
Linear	26.17	0.853	29.28	19.11	0.597	78.81	22.73	0.675	103.21
Cubic	25.71	0.842	31.13	17.97	0.533	91.93	21.65	0.618	119.41
ACVTT	27.34	0.882	25.52	20.19	0.662	71.55	23.47	0.728	95.44
Res-SRDiff	28.04	0.882	24.19	21.67	0.691	56.34	21.15	0.554	153.02
ArSSR	27.25	0.877	25.55	20.31	0.657	65.99	23.16	0.709	96.49
CycleINR-style	26.17	0.853	29.28	20.32	0.658	66.10	22.90	0.693	101.10
ResUNet	27.46	0.881	25.24	22.25	0.737	53.57	23.38	0.721	97.04
Pix2Pix-GAN	27.11	0.851	27.02	21.91	0.729	53.26	22.45	0.711	105.81
SwinIR	28.65	0.900	22.46	22.36	0.749	50.87	23.95	0.752	85.97
SAINT-style	28.67	0.897	22.84	23.26	0.770	46.44	23.84	0.742	89.41
I3Net	31.31	0.928	17.53	24.01	0.792	41.01	23.49	0.721	93.57
HAT	28.72	0.902	22.25	23.09	0.769	47.25	23.95	0.752	86.14
SMamba-UNet	29.35	0.908	21.14	22.73	0.762	49.45	23.91	0.750	86.82

Locally adapted implementations retain their recorded training budgets; the master uses extended HAT training on fastMRI. This is not an equal-compute leaderboard. Standard and longer-budget results are reported separately.

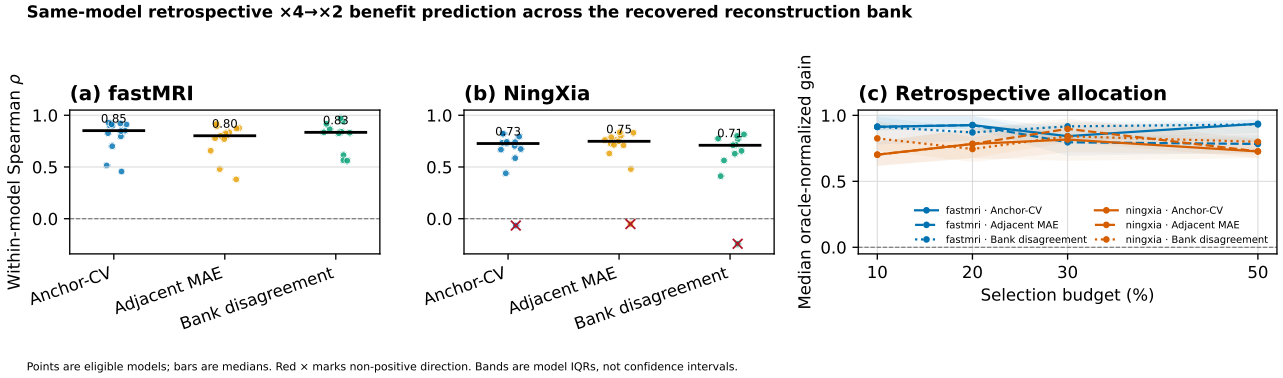


Figure 6. Fixed-rule retrospective same-model consequence, moved intact from the main paper. All eligible models remain; red crosses preserve adverse directions. Model-IQR bands are not confidence intervals. Factor-specific checkpoints can differ, and controlling current error weakens the benefit association.

tables and this native-metric ledger remain distinct contracts. Automatic threshold-mask Dice/HD95 columns are not interpreted as anatomical segmentation.

All model names denote the local implementations, including generic SR models adapted to 2.5-D conditioning, inherited GAN, and local SAINT/CycleINR variants. The recorded master profiles include different training budgets and early stopping; no equal-compute claim is made. PASTNet's fastMRI development trajectory remains development evidence rather than a new independent three-cohort baseline result.

What the full bank adds. Using all 13 native-metric baselines gives 78 model pairs per cohort. Median case-error rank correlations are .997 on IXI, .899 on fastMRI, and 1.000 on eight NingXia patient folds. These descriptive estimates show shared difficulty across the full bank while retaining the lower fastMRI association than in the masked analysis. They do not imply independent model pairs or extend the five-model intervention to 13 models.

Budget protocol and negative boundaries. For every recorded 60/90-epoch experiment, compare its matched standard-profile cases; use 10,000 paired bootstrap draws (seed 20260910), examinations for fastMRI and patient folds for NingXia. The primary MAE intervals simultaneously cover all 20 comparisons using the bootstrap maximum standardized deviation. All five metric changes remain in the

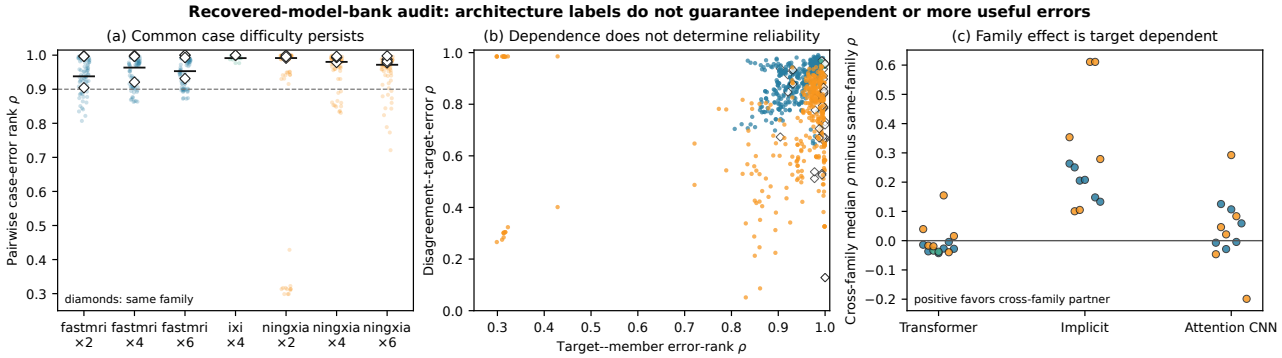


Figure 7. Exploratory recovered-model-bank audit. (a) Case difficulty remains shared across architectures; diamonds denote same-family pairs. (b) Pairwise error dependence does not uniquely determine how disagreement ranks target error. (c) Cross-family minus matched same-family reliability can be positive or negative. Orange denotes NingXia, including the retained degraded Res-SRDiff boundary.

Table 17. Architecture-family same-target result on 24 fastMRI examinations at ×4. Observable gain is disjoint minus clean shared fraction. Ranking intervals resample examinations.

Model	Family	Observable gain [95% CI]	$\rho_{\text{shared}} \rightarrow \rho_{\text{disjoint}}$	Ranking $\Delta\rho$ [95% CI]
HAT	Transformer SR	.178 [.170,.187]	.896→.952	.057 [−.030,.186]
SwinIR	Transformer SR	.176 [.164,.188]	.883→.945	.062 [−.051,.234]
I3Net	Inter/intra-slice CNN	.080 [.072,.088]	.943→.961	.018 [−.061,.107]
SAINT	Spatial conditional CNN	.137 [.125,.149]	.934→.924	−.010 [−.143,.114]
CycleINR	Implicit coordinate	.226 [.219,.233]	.634→.976	.342 [.113,.711]

numerical record. Fifteen MAE intervals favor longer training; five cross zero. Runs are separate training experiments, not a controlled single-checkpoint continuation, and standard values are never replaced by their best budget.

NingXia Res-SRDiff ×2 MAE changes from 440.61 at 30 epochs to 140.45 at 60 and 118.25 at 90, still above Linear's 78.53. A short-budget failure therefore cannot establish an intrinsic diffusion-family failure. Conversely, CycleINR ×2 barely changes (.000006 MAE reduction), and five simultaneous intervals remain inconclusive. The source note that fastMRI ACVTT ×6 worsened at 60 epochs is contradicted by its numbers: PSNR improves 19.8309 to 20.0269, and MAE 72.4367 to 70.9050; all five metric directions improve. The original note is preserved and this correction is explicit.

4.15. Preserved common-support reconstruction benchmark

Table 18. Frozen IXI $\times 4$ replication and reflection-matched identity swap. Effects are disjoint minus shared observable fraction. Clean and balanced effects are examination medians; mirror difference is residue-1 minus residue-3. All intervals resample examinations.

Model	Clean gain [95% CI]	Ranking $\Delta\rho$ [95% CI]	Reflection-balanced gain [95% CI]	Mirror difference [95% CI]
HAT	.290 [.286,.296]	.031 [.016,.053]	.292 [.286,.297]	.000 [−.005,.007]
SwinIR	.302 [.295,.307]	.035 [.019,.059]	.300 [.295,.308]	.001 [−.003,.006]
I3Net	.252 [.244,.260]	.025 [.008,.053]	.251 [.244,.257]	.002 [−.004,.012]
SAINT-style	.298 [.292,.305]	.032 [.017,.055]	.297 [.288,.302]	−.0003 [−.006,.006]
CycleINR-style	.296 [.291,.300]	.134 [.058,.258]	.295 [.290,.299]	.001 [−.0004,.004]

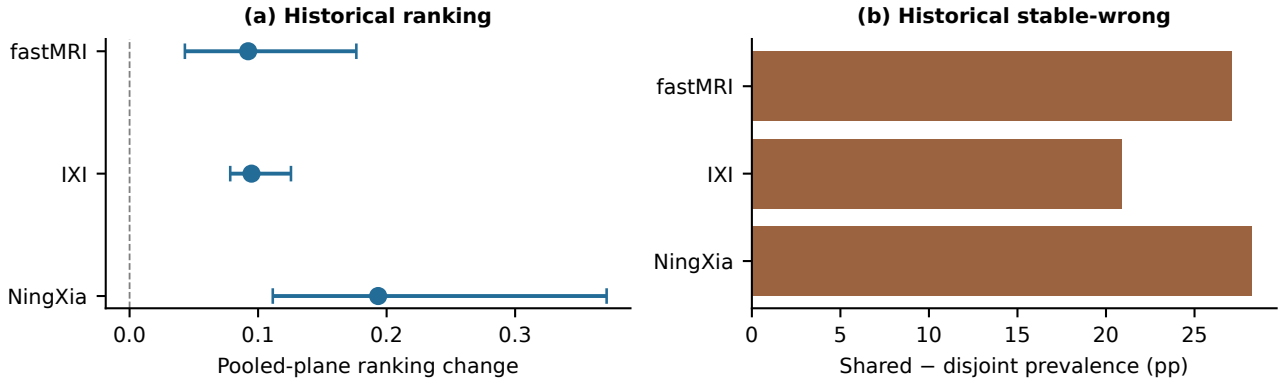


Figure 8. Retained historical pooled-plane disagreement ranking and stable-wrong prevalence. Ranking uses fixed phase-0 MAE on fastMRI/NingXia and disjoint-pair MAE on IXI. Intervals resample examination/patient clusters; prevalence uses the original test-relative thresholds. Neither is an examination-level point correlation or calibrated event rate.

Table 19. Original clean shared/disjoint comparison. Ranking against a fixed endpoint (phase-0 MAE on fastMRI/NingXia; disjoint-pair mean MAE on IXI) pools plane/model observations; intervals resample examinations/patients. Stable-wrong thresholds are test-relative, fixed from the disjoint condition.

Cohort	Independent units	ρ_{disjoint}	ρ_{shared}	$\Delta\rho$ [95% CI]	Observable error / stable-wrong Δ
fastMRI	24 examinations	.959	.867	.092 [.043,.176]	2.61% / +27.1 pp
IXI	115 examinations	.981	.886	.095 [.078,.126]	1.58% / +20.9 pp
NingXia	8 patients (22 series)	.892	.698	.193 [.111,.371]	.25% / +28.2 pp

Table 20. Clean same-target dependence control. Stable-wrong Δ uses high-error/low-disagreement thresholds frozen from the zero-overlap condition.

Cohort	Units	ρ_{disjoint}	ρ_{shared}	$\Delta\rho$ [95% CI]	Shared / observable error
fastMRI	24 examinations	.959	.867	.092 [.043,.176]	97.39% / 2.61%
IXI	115 examinations	.981	.886	.095 [.078,.126]	98.42% / 1.58%
NingXia	8 patients, 22 series	.892	.698	.193 [.111,.371]	99.75% / .25%

Table 21. Analytical shared influence versus count. Values are Spearman association with error alignment.

Cohort	Count	Common coefficient mass	Gain
fastMRI	.503	.815	.312
IXI	.466	.789	.323
NingXia	.630	.855	.225

Table 22. Broad neural context grid. ρ_c uses overlap count and ρ_I neural shared influence. Advantage intervals are clustered.

Cohort	Model	ρ_c align.	ρ_I align.	Adv. [95% CI]	ρ_I shared	ρ_I observable	partial ρ_I align.
fastMRI	HAT	.799	.838	.039 [.028,.049]	.829	−.830	.359
	SwinIR	.796	.851	.055 [.048,.061]	.845	−.845	.395
IXI	HAT	.714	.797	.082 [.076,.088]	.794	−.795	.315
	SwinIR	.712	.787	.075 [.070,.080]	.786	−.787	.289
NingXia	HAT	.743	.786	.043 [.027,.057]	.792	−.792	.235
	SwinIR	.735	.772	.036 [.016,.057]	.781	−.781	.197

Table 23. Neural influence partial Spearman at exactly one shared anchor, controlling span, asymmetry, and examination. Error-alignment intervals are shown.

Cohort	Model	Error alignment	Shared fraction	Observable fraction	Disagreement
fastMRI	HAT	.169 [.053,.283]	.135	−.135	−.161
	SwinIR	.350 [.212,.464]	.321	−.321	−.286
IXI	HAT	.346 [.259,.424]	.360	−.360	−.543
	SwinIR	.258 [.181,.335]	.241	−.241	−.587
NingXia	HAT	.014 [−.232,.223]	−.006	.006	−.102
	SwinIR	−.039 [−.175,.144]	−.039	.039	−.072

Table 24. Partial Spearman ranges across HAT/SwinIR. Local arrows are supported in every stratum. "Ranking" counts architecture intervals excluding zero.

Cohort	Influence→alignment	Alignment→norm. disagreement	Influence→observable	Alignment→ranking
fastMRI	.372–.410	−.933–−.926	−.412–−.370	−.333–−.143 (0/2)
IXI	.321–.347	−.893–−.890	−.342–−.298	−.643–−.500 (2/2)
NingXia	.219–.258	−.898–−.889	−.231–−.211	−.667–−.548 (1/2)

Table 25. Examination-level $\times 4$ difficulty ranking. Lower nAURC is better.

Cohort	Method	Spearman	Kendall	nAURC
fastMRI	Anchor-CV / adjacent MAE / adjacent RMSE	.943 / .934 / .942	.812 / .790 / .804	.0231 / .0222 / .0241
IXI	Anchor-CV / adjacent MAE / adjacent RMSE	.955 / .956 / .956	.831 / .834 / .826	.0150 / .0145 / .0147
NingXia	Anchor-CV / adjacent MAE / adjacent RMSE	.958 / .973 / .959	.844 / .879 / .835	.0112 / .0072 / .0123

Table 26. Current-error sensitivity, with every eligible model retained. Entries are model-median correlations; the final column counts positive pointwise partial-correlation intervals. This is a diagnostic adjustment using retrospective target error, not a deployable score.

Cohort	Score	Raw ρ	Partial $\rho \mid E_4$	Positive partial CI / models
fastMRI	Anchor-CV	.851	.203	0/13
	Adjacent MAE	.802	−.068	0/13
	Bank disagreement	.835	.056	1/11
NingXia	Anchor-CV	.727	.291	0/13
	Adjacent MAE	.748	.426	1/13
	Bank disagreement	.710	.389	0/11

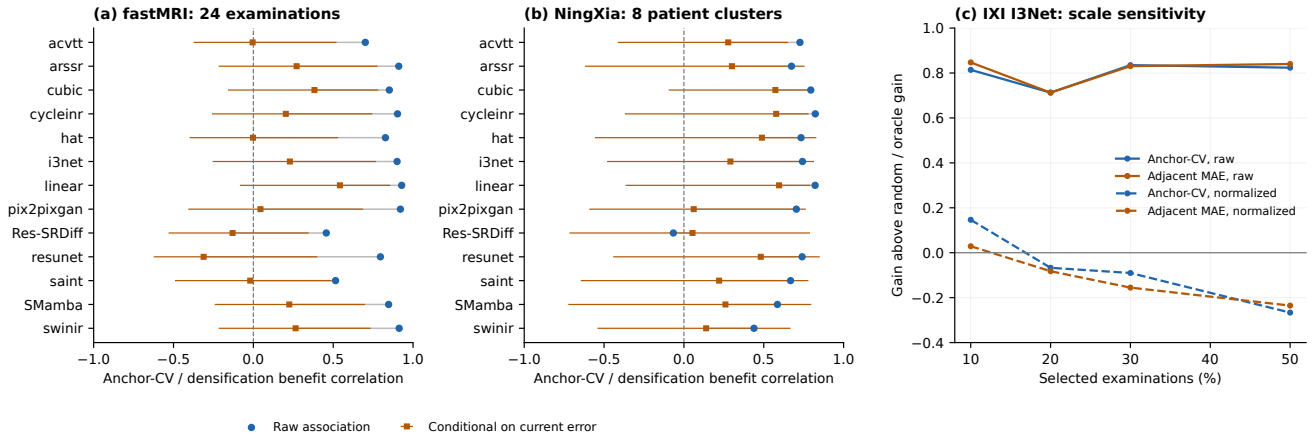


Figure 9. Benefit boundaries. (a,b) Raw Anchor-CV associations (blue) and associations conditional on current error (orange), with patient/examination-bootstrap 95% intervals for the latter. All 26 conditional intervals cross zero. (c) Recovered IXI I3Net allocation shows a reversal after acquired-range normalization against the same raw-scale benefit endpoint. Lines are point estimates, not uncertainty bands.

Table 27. IXI $\times 4$: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	26.1722	0.8529	29.2781	6553.8673	0.9390
Cubic	25.7138	0.8422	31.1269	7217.2979	0.9303
ACVTT	27.3431	0.8822	25.5217	5206.8325	0.9557
Res-SRDiff	28.0441	0.8821	24.1938	4664.4129	0.9611
ArSSR	27.2467	0.8773	25.5509	5267.3658	0.9533
CycleINR-style	26.1728	0.8527	29.2765	6553.8469	0.9390
ResUNet	27.4611	0.8812	25.2395	4985.4989	0.9552
Pix2Pix-GAN	27.1070	0.8506	27.0202	5372.2668	0.9538
SwinIR	28.6485	0.8997	22.4556	4137.5109	0.9658
SAINT-style	28.6677	0.8970	22.8365	4128.0151	0.9660
I3Net	31.3058	0.9285	17.5276	2565.3177	0.9788
HAT	28.7249	0.9016	22.2500	4088.7422	0.9663
SMamba-UNet	29.3536	0.9077	21.1352	3691.2381	0.9699

Table 28. fastMRI $\times 4$: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	19.1133	0.5970	78.8111	23092.0046	0.7733
Cubic	17.9736	0.5332	91.9330	29740.2642	0.7285
ACVTT	20.1941	0.6624	71.5480	18286.6190	0.8320
Res-SRDiff	21.6727	0.6909	56.3371	13095.4191	0.8814
ArSSR	20.3143	0.6575	65.9931	17745.8948	0.8301
CycleINR-style	20.3156	0.6577	66.0985	17743.2358	0.8302
ResUNet	22.2549	0.7374	53.5705	11842.8147	0.8955
Pix2Pix-GAN	21.9090	0.7294	53.2622	12485.1668	0.8915
SwinIR	22.3578	0.7494	50.8733	11636.3791	0.8977
SAINT-style	23.2605	0.7705	46.4417	9347.6662	0.9177
I3Net	24.0102	0.7921	41.0101	7755.6546	0.9308
HAT	23.0894	0.7688	47.2467	9773.1765	0.9138
SMamba-UNet	22.7322	0.7618	49.4487	10826.4468	0.9051

Table 29. NingXia $\times 4$: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	22.7335	0.6749	103.2052	42189.1127	0.8576
Cubic	21.6460	0.6184	119.4078	53313.5368	0.8252
ACVTT	23.4735	0.7281	95.4376	35779.4840	0.8804
Res-SRDiff	21.1538	0.5538	153.0213	66533.1526	0.8207
ArSSR	23.1634	0.7091	96.4890	38343.9409	0.8708
CycleINR-style	22.9009	0.6935	101.0990	40735.3568	0.8625
ResUNet	23.3816	0.7209	97.0409	36316.2290	0.8785
Pix2Pix-GAN	22.4530	0.7112	105.8119	46398.6707	0.8552
SwinIR	23.9513	0.7525	85.9679	32030.5455	0.8930
SAINT-style	23.8350	0.7422	89.4140	32992.9496	0.8905
I3Net	23.4911	0.7208	93.5655	35725.4805	0.8825
HAT	23.9499	0.7519	86.1449	32226.9860	0.8933
SMamba-UNet	23.9126	0.7500	86.8152	32498.3175	0.8917

Table 30. fastMRI $\times 2$: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	21.6348	0.7030	53.9754	11591.9388	0.8732
Cubic	20.3321	0.6363	64.1081	15236.6373	0.8370
ACVTT	21.7910	0.7110	53.4000	11160.0709	0.8783
Res-SRDiff	22.4766	0.6511	50.3378	9901.7857	0.9034
ArSSR	23.1137	0.7613	43.5812	8391.3334	0.9140
CycleINR-style	23.0957	0.7593	43.7063	8411.5115	0.9132
ResUNet	24.1700	0.7913	39.7905	6909.6572	0.9338
Pix2Pix-GAN	26.0734	0.8302	31.2304	4584.5238	0.9565
SwinIR	24.8716	0.8153	35.9392	5990.3711	0.9428
SAINT-style	24.4308	0.7809	39.6861	6749.7212	0.9373
I3Net	27.7711	0.8670	25.3325	3113.5377	0.9700
HAT	24.6475	0.8106	37.2729	6225.7429	0.9409
SMamba-UNet	24.6548	0.8077	37.1110	6361.0863	0.9401

Table 31. fastMRI $\times 6$: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	18.2648	0.5595	88.2647	28258.4309	0.7215
Cubic	18.2648	0.5595	88.2647	28258.4309	0.7215
ACVTT	19.8309	0.6601	72.4367	20031.3896	0.8120
Res-SRDiff	21.2529	0.6791	59.4388	14747.2531	0.8702
ArSSR	19.4214	0.6156	74.9270	21914.5364	0.7872
CycleINR-style	19.3873	0.6158	74.9017	22086.6021	0.7854
ResUNet	22.1434	0.7375	53.8609	12277.0333	0.8921
Pix2Pix-GAN	21.6510	0.7250	54.8359	13567.5373	0.8828
SwinIR	21.8398	0.7305	56.4490	13267.8010	0.8840
SAINT-style	22.3172	0.7466	52.5303	11683.3596	0.8979
I3Net	22.8308	0.7617	47.0732	10309.8900	0.9088
HAT	21.8230	0.7313	55.7775	13341.4608	0.8829
SMamba-UNet	21.9153	0.7313	56.5002	13166.6662	0.8848

Table 32. NingXia $\times 2$: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	24.9741	0.7264	78.5349	27551.4768	0.9112
Cubic	24.1081	0.6767	89.5215	33708.1987	0.8952
ACVTT	25.3139	0.7518	76.2396	25631.8031	0.9180
Res-SRDiff	15.3187	0.2945	440.6059	311894.3839	0.8320
ArSSR	25.1538	0.7427	76.6719	26571.4634	0.9146
CycleINR-style	25.1538	0.7427	76.6723	26571.4525	0.9146
ResUNet	25.2629	0.7478	76.6937	25684.3115	0.9170
Pix2Pix-GAN	25.2116	0.7543	74.8168	25448.9329	0.9169
SwinIR	25.5845	0.7651	73.1350	24249.0709	0.9226
SAINT-style	25.3398	0.7535	75.9435	25247.7430	0.9191
I3Net	25.7679	0.7574	70.8835	23483.9767	0.9265
HAT	25.8489	0.7738	70.0441	22767.1915	0.9276
SMamba-UNet	25.4409	0.7603	74.6767	24857.0321	0.9201

Table 33. NingXia $\times 6$: complete native metrics, recorded master profile.

Method	PSNR	SSIM	MAE	MSE	NCC
Linear	21.7375	0.6509	118.6269	53812.4881	0.8146
Cubic	21.6523	0.6483	119.6982	54616.6079	0.8118
ACVTT	22.6160	0.7126	108.2221	44622.4723	0.8490
Res-SRDiff	20.4586	0.5616	156.7453	75246.6372	0.7890
ArSSR	22.3530	0.7007	108.1897	47147.2527	0.8389
CycleINR-style	21.8961	0.6728	116.1079	52220.1671	0.8214
ResUNet	22.7683	0.7143	104.7535	42861.1183	0.8551
Pix2Pix-GAN	21.1353	0.6662	129.1480	61058.8712	0.8005
SwinIR	22.9792	0.7362	98.5621	40916.0806	0.8624
SAINT-style	22.7341	0.7233	106.0398	44729.0092	0.8562
I3Net	22.3951	0.6940	111.4996	46546.2263	0.8452
HAT	23.0110	0.7361	98.5351	40588.4310	0.8635
SMamba-UNet	22.9507	0.7336	100.4715	41414.3824	0.8611

Table 34. All recorded extended-budget results. Native five-metric values are kept separately from the master-profile ranking.

Cohort	Method	Factor	Epochs	PSNR	SSIM	MAE	MSE	NCC
NX	ArSSR	4	60	23.4566	0.7241	91.8598	36115.4376	0.8793
NX	ResUNet	4	60	23.7562	0.7379	90.5450	33771.8800	0.8876
NX	SAINT-style	4	60	23.8388	0.7473	88.1769	33105.9966	0.8909
NX	SMamba-UNet	4	60	24.0160	0.7542	84.8084	31545.2244	0.8947
NX	SwinIR	4	60	24.0370	0.7573	83.8378	31317.4826	0.8946
NX	CycleINR-style	2	60	25.1538	0.7427	76.6723	26571.4690	0.9146
NX	HAT	2	60	26.1387	0.7853	66.2919	21520.2802	0.9321
NX	Res-SRDiff	2	60	22.6714	0.5619	140.4503	56206.7760	0.8894
NX	SAINT-style	2	60	25.6706	0.7677	72.4046	23699.0011	0.9245
NX	SMamba-UNet	2	60	25.9101	0.7813	68.4655	22757.3826	0.9281
NX	SwinIR	2	60	26.0556	0.7821	67.3308	21930.8546	0.9310
NX	ArSSR	6	60	22.4720	0.7020	106.4477	46384.2568	0.8423
NX	CycleINR-style	6	60	22.1578	0.6886	111.6263	49569.1804	0.8335
NX	Res-SRDiff	6	60	20.8328	0.6008	139.9794	62636.9837	0.7782
NX	ResUNet	6	60	22.9161	0.7281	100.4640	41591.4635	0.8610
NX	SwinIR	6	60	23.0201	0.7392	97.7710	40687.7343	0.8637
NX	Res-SRDiff	2	90	23.5044	0.6178	118.2457	43546.3681	0.8924
fastMRI	SwinIR	2	60	25.4180	0.8274	33.7890	5326.9072	0.9500
fastMRI	ACVTT	6	60	20.0269	0.6677	70.9050	19173.6891	0.8211
fastMRI	Res-SRDiff	2	60	24.0661	0.7540	40.8709	7086.3766	0.9320

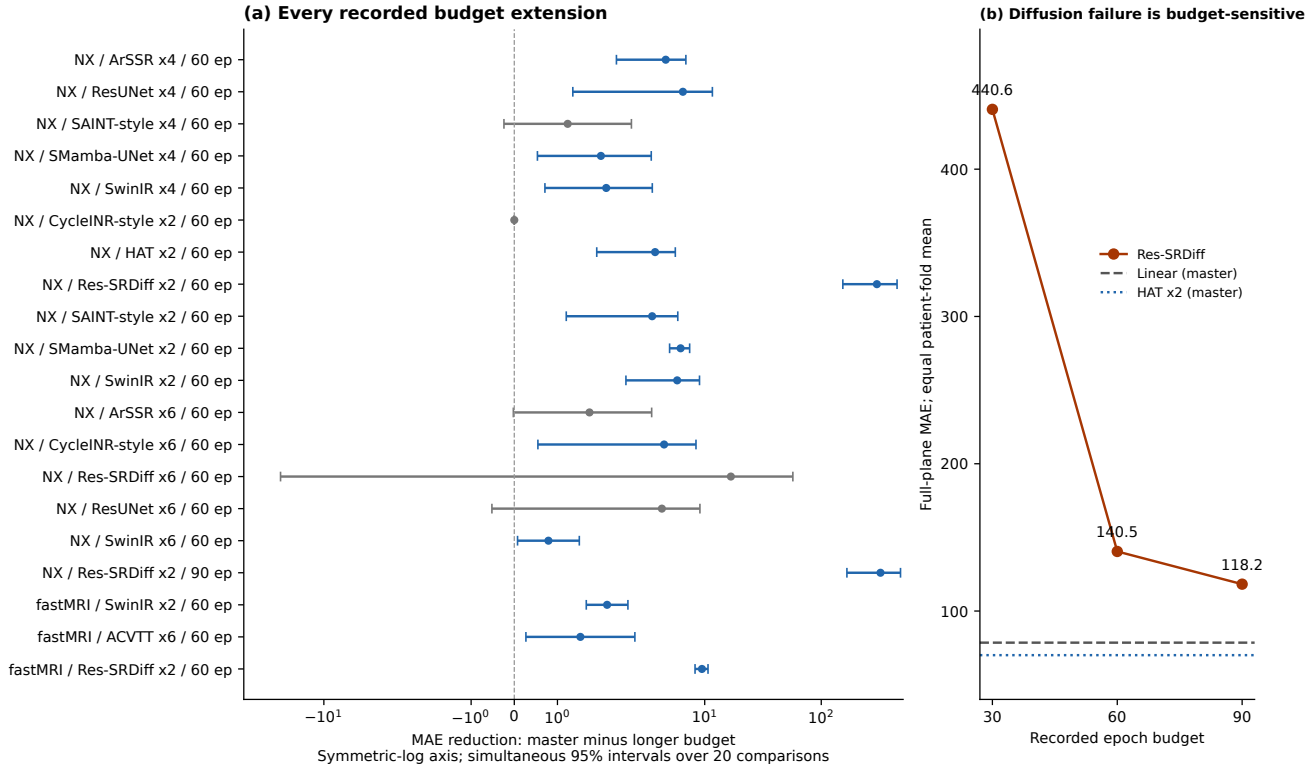


Figure 10. All 20 recorded budget extensions. Positive reductions favor longer training; the symmetric-log axis keeps the large diffusion changes and small effects visible. Intervals simultaneously cover the 20 primary MAE contrasts. Five cross zero. Budget sensitivity does not validate a new QA score or erase the original short-budget failure.

Table 35. fastMRI recovered reconstruction context. Four metrics at every available factor; "— denotes unavailable audited cells.

	×2				×4				×6			
Method	MAE	PSNR	SSIM	NCC	MAE	PSNR	SSIM	NCC	MAE	PSNR	SSIM	NCC
Linear	98.66	16.77	0.7707	0.7816	134.54	14.24	0.5884	0.5953	148.66	13.42	0.5042	0.5098
Cubic	114.51	15.63	0.7432	0.7414	153.24	13.21	0.5443	0.5411	148.66	13.42	0.5042	0.5098
ACSI-Net	—	—	—	—	—	—	—	—	—	—	—	—
ACVTT	97.73	16.94	0.7794	0.7911	122.23	15.38	0.6676	0.6886	124.63	15.00	0.6264	0.6485
ArSSR	80.03	18.19	0.8342	0.8506	114.21	15.38	0.6659	0.6888	128.03	14.51	0.5814	0.6057
CycleINR	80.25	18.17	0.8346	0.8496	114.39	15.38	0.6659	0.6888	128.25	14.47	0.5774	0.6033
HAT	68.03	19.59	0.8909	0.8940	86.09	17.80	0.8277	0.8327	96.75	16.82	0.7727	0.7826
I3Net	45.61	22.57	0.9459	0.9473	70.87	19.03	0.8714	0.8757	81.61	17.82	0.8256	0.8333
Pix2PixGAN	56.64	20.91	0.9204	0.9220	92.41	16.95	0.8076	0.8067	95.61	16.61	0.7913	0.7911
Res-SRDiff	89.32	17.49	0.8203	0.8272	95.98	16.75	0.7781	0.7865	102.11	16.22	0.7655	0.7655
ResUNet	72.73	19.12	0.8785	0.8814	92.68	17.27	0.8014	0.8087	92.93	17.14	0.7940	0.8013
SAINT	72.08	19.31	0.8857	0.8874	80.14	18.30	0.8477	0.8511	90.87	17.35	0.8071	0.8116
SMamba-UNet	67.94	19.52	0.8887	0.8916	85.88	17.68	0.8194	0.8267	97.94	16.86	0.7741	0.7849
SwinIR	65.69	19.78	0.8948	0.8978	90.37	17.37	0.8028	0.8121	97.73	16.83	0.7725	0.7835

Table 36. IXI recovered reconstruction context. Four metrics at every available factor; "— denotes unavailable audited cells.

	×2				×4				×6			
Method	MAE	PSNR	SSIM	NCC	MAE	PSNR	SSIM	NCC	MAE	PSNR	SSIM	NCC
Linear	—	—	—	—	65.99	20.34	0.9003	0.9030	—	—	—	—
Cubic	—	—	—	—	69.87	19.99	0.8984	0.8977	—	—	—	—
ACSI-Net	—	—	—	—	47.13	23.12	0.9480	0.9482	—	—	—	—
ACVTT	—	—	—	—	—	—	—	—	—	—	—	—
ArSSR	—	—	—	—	—	—	—	—	—	—	—	—
CycleINR	—	—	—	—	—	—	—	—	—	—	—	—
HAT	—	—	—	—	50.18	22.51	0.9414	0.9413	—	—	—	—
I3Net	—	—	—	—	39.26	25.03	0.9645	0.9644	—	—	—	—
Pix2PixGAN	—	—	—	—	—	—	—	—	—	—	—	—
Res-SRDiff	—	—	—	—	—	—	—	—	—	—	—	—
ResUNet	—	—	—	—	—	—	—	—	—	—	—	—
SAINT	—	—	—	—	—	—	—	—	—	—	—	—
SMamba-UNet	—	—	—	—	—	—	—	—	—	—	—	—
SwinIR	—	—	—	—	50.55	22.46	0.9406	0.9404	—	—	—	—

Table 37. NingXia recovered reconstruction context. Four metrics at every available factor; "— denotes unavailable audited cells.

	×2				×4				×6			
Method	MAE	PSNR	SSIM	NCC	MAE	PSNR	SSIM	NCC	MAE	PSNR	SSIM	NCC
Linear	136.78	18.12	0.7611	0.7697	175.93	16.14	0.6350	0.6408	201.72	15.02	0.5343	0.5378
Cubic	153.41	17.29	0.7423	0.7412	197.41	15.20	0.5998	0.5968	204.51	14.92	0.5298	0.5329
ACSI-Net	—	—	—	—	—	—	—	—	—	—	—	—
ACVTT	131.78	18.45	0.7711	0.7850	162.77	16.85	0.6640	0.6832	185.36	15.89	0.5626	0.5881
ArSSR	133.56	18.28	0.7662	0.7773	165.77	16.51	0.6509	0.6655	186.41	15.57	0.5523	0.5704
CycleINR	133.56	18.28	0.7662	0.7773	172.33	16.30	0.6417	0.6501	198.16	15.15	0.5400	0.5466
HAT	122.16	18.91	0.7963	0.8082	149.41	17.25	0.6961	0.7153	172.24	16.23	0.5903	0.6229
I3Net	124.09	18.82	0.7860	0.8046	162.62	16.80	0.6332	0.6836	188.46	15.65	0.5102	0.5723
Pix2PixGAN	129.67	18.35	0.7744	0.7855	183.70	15.73	0.6094	0.6177	218.87	14.43	0.4596	0.4733
Res-SRDiff	479.32	10.40	0.5459	0.6322	225.44	14.93	0.5519	0.5661	254.89	13.91	0.4379	0.4685
ResUNet	132.33	18.40	0.7697	0.7828	164.52	16.77	0.6620	0.6785	180.15	16.01	0.5812	0.6030
SAINT	130.89	18.48	0.7742	0.7876	154.55	17.15	0.6904	0.7069	186.94	15.89	0.5670	0.5960
SMamba-UNet	129.36	18.57	0.7764	0.7906	150.46	17.22	0.6868	0.7091	176.48	16.17	0.5814	0.6161
SwinIR	127.41	18.67	0.7829	0.7961	148.66	17.27	0.6975	0.7152	172.19	16.20	0.5853	0.6202

5. Post-hoc Submission Sensitivities

The accepted primary evidence and its raw arrays were hashed before these analyses. These are post-hoc sensitivities, not prespecified primary tests. All original targets and frozen predictions are retained; no network is rerun, no new mask threshold is searched and no new score is formed. Each examination has equal weight after target averaging. We use 5,000 common examination-bootstrap draws for all 20 ordered pairs within each cohort. Simultaneous intervals apply separately to each cohort/support/geometry endpoint, not jointly to the entire sensitivity collection. Model pairs share examinations and architectures, and are not independent replications.

5.1. Support contracts

The original target 1%-foreground mask, the existing union of the nearest two acquired phase-0 foreground masks, and all common finite image pixels are the only contracts. Each acquired anchor is thresholded at 1% of its own maximum absolute intensity. Support is identical for all pairs and both conditions at the same target; the acquired mask never accesses the target. Shared O, disjoint O, their difference and pixel counts for every pair are retained in the accompanying numerical files. The target remains necessary for retrospective error evaluation under every contract.

All 20 effects and all 20 simultaneous intervals per cohort remain positive under each support. The maximum pair-effect rank movement is one on fastMRI and zero on IXI; minimum Spearman agreement with the original pair ordering is .997. This rules against dependence on the tested target-derived support, not against all conceivable support effects or against shared measurement/geometry confounding.

5.2. Residue, mirror and nearest-distance checks

Targets are divided by their physical index modulo four. Within each examination we average each residue class, then weight residues 1 and 3 equally. The mirror difference is the residue-1 effect minus the residue-3 effect, analogous to the historical reflection-class analysis but retaining this experiment's mean estimand. It is not a new reflected-image prediction or an identity-only counterfactual. Nearest-distance matching requires actual bracketing distances $\{1, 3\}$ in both phases. All originally selected targets satisfy this gate, so the matched balanced estimate equals the residue-balanced estimate.

The balanced local conclusion survives in both cohorts, as does each residue separately. However, mirror intervals exclude zero for eight fastMRI pairs (two positive, six negative) and six IXI pairs (two positive, four negative). These are retained orientation warnings; the remaining intervals do not prove equivalence. The stricter complete four-anchor contract requires two actual acquired anchors on each side in both phases and both residues within an examination. All IXI selected targets meet it. For fastMRI only target $z = 7$ meets the complete-context requirement, leaving no residue-1 target: every such balanced pair estimate is NA. Boundary anchors are never duplicated or synthesized.

5.3. Joint family robustness

The complete matrices remain primary. The following descriptive summaries resample whole examinations jointly across all pair columns; they do not resample pairs as independent units. Pair-family mean/median is recomputed inside each bootstrap sample. Excluding one architecture removes every pair involving it in either direction, leaving twelve. All family intervals remain positive, including after removing each individual architecture. These results test dominance within the observed model family, not generalization to arbitrary architectures.

Table 38. fastmri: all 20 post-hoc support contrasts. ΔO [simultaneous 95% CI], percentage points.

Ordered pair	Target mask	Acquired union	Full image
CycleINR→HAT	11.20 [9.46,12.95]	11.07 [9.33,12.81]	11.07 [9.33,12.81]
CycleINR→I3Net	6.78 [5.39,8.18]	6.74 [5.36,8.12]	6.74 [5.36,8.12]
CycleINR→SAINT	7.58 [5.76,9.40]	7.49 [5.68,9.30]	7.49 [5.68,9.30]
CycleINR→SwinIR	11.49 [9.39,13.59]	11.36 [9.27,13.45]	11.35 [9.25,13.44]
HAT→CycleINR	11.56 [10.34,12.78]	11.43 [10.22,12.65]	11.41 [10.19,12.64]
HAT→I3Net	9.76 [8.75,10.76]	9.71 [8.72,10.71]	9.71 [8.71,10.70]
HAT→SAINT	12.96 [11.60,14.32]	12.87 [11.51,14.23]	12.86 [11.50,14.21]
HAT→SwinIR	16.02 [14.61,17.44]	15.88 [14.46,17.30]	15.87 [14.45,17.29]
I3Net→CycleINR	6.24 [5.23,7.24]	6.16 [5.16,7.17]	6.15 [5.15,7.16]
I3Net→HAT	8.89 [7.64,10.15]	8.82 [7.57,10.07]	8.82 [7.57,10.06]
I3Net→SAINT	10.70 [9.73,11.67]	10.63 [9.66,11.60]	10.62 [9.65,11.59]
I3Net→SwinIR	7.75 [6.42,9.08]	7.68 [6.36,8.99]	7.67 [6.36,8.99]
SAINT→CycleINR	7.77 [6.49,9.04]	7.68 [6.40,8.95]	7.66 [6.39,8.94]
SAINT→HAT	12.65 [11.25,14.06]	12.57 [11.17,13.97]	12.56 [11.16,13.96]
SAINT→I3Net	11.13 [10.08,12.17]	11.10 [10.06,12.13]	11.08 [10.05,12.12]
SAINT→SwinIR	10.85 [9.24,12.45]	10.77 [9.18,12.37]	10.76 [9.16,12.35]
SwinIR→CycleINR	11.06 [9.38,12.74]	10.94 [9.26,12.61]	10.91 [9.23,12.59]
SwinIR→HAT	15.67 [14.32,17.03]	15.54 [14.18,16.89]	15.53 [14.17,16.88]
SwinIR→I3Net	8.65 [7.45,9.84]	8.61 [7.44,9.79]	8.60 [7.43,9.78]
SwinIR→SAINT	10.85 [9.25,12.45]	10.77 [9.18,12.37]	10.76 [9.17,12.35]

Table 39. ixi: all 20 post-hoc support contrasts. ΔO [simultaneous 95% CI], percentage points.

Ordered pair	Target mask	Acquired union	Full image
CycleINR→HAT	21.12 [20.56,21.69]	20.99 [20.42,21.55]	20.96 [20.39,21.52]
CycleINR→I3Net	13.15 [12.22,14.08]	13.07 [12.13,14.01]	13.04 [12.12,13.97]
CycleINR→SAINT	22.55 [21.99,23.11]	22.40 [21.84,22.96]	22.35 [21.80,22.91]
CycleINR→SwinIR	22.33 [21.79,22.88]	22.19 [21.64,22.73]	22.15 [21.61,22.69]
HAT→CycleINR	21.20 [20.58,21.81]	21.05 [20.43,21.66]	21.02 [20.41,21.63]
HAT→I3Net	18.33 [17.32,19.34]	18.27 [17.26,19.28]	18.22 [17.21,19.22]
HAT→SAINT	27.55 [26.85,28.25]	27.42 [26.72,28.12]	27.35 [26.66,28.05]
HAT→SwinIR	28.53 [27.83,29.24]	28.42 [27.71,29.12]	28.36 [27.66,29.05]
I3Net→CycleINR	13.11 [12.09,14.13]	13.01 [11.99,14.03]	13.00 [11.98,14.01]
I3Net→HAT	18.00 [16.90,19.10]	17.93 [16.83,19.03]	17.88 [16.79,18.97]
I3Net→SAINT	17.95 [16.84,19.07]	17.87 [16.75,18.98]	17.81 [16.71,18.92]
I3Net→SwinIR	17.96 [16.87,19.06]	17.89 [16.79,18.98]	17.84 [16.75,18.92]
SAINT→CycleINR	22.59 [21.99,23.20]	22.43 [21.82,23.03]	22.38 [21.79,22.98]
SAINT→HAT	27.48 [26.79,28.18]	27.36 [26.66,28.06]	27.29 [26.60,27.98]
SAINT→I3Net	18.24 [17.21,19.27]	18.16 [17.13,19.19]	18.11 [17.08,19.13]
SAINT→SwinIR	28.10 [27.40,28.80]	27.97 [27.26,28.67]	27.90 [27.20,28.59]
SwinIR→CycleINR	22.40 [21.80,23.01]	22.24 [21.64,22.84]	22.20 [21.61,22.80]
SwinIR→HAT	28.49 [27.77,29.20]	28.38 [27.66,29.09]	28.31 [27.61,29.02]
SwinIR→I3Net	18.27 [17.25,19.28]	18.20 [17.18,19.22]	18.15 [17.14,19.16]
SwinIR→SAINT	28.13 [27.42,28.84]	27.99 [27.28,28.71]	27.93 [27.22,28.63]

Table 40. fastmri: all ordered-pair geometry contrasts, percentage points, with simultaneous 95% intervals per column. Mirror difference is an orientation diagnostic, not an observability-gain endpoint.

Ordered pair	Residue 1	Residue 3	Balanced	Mirror difference
CycleINR→HAT	10.00 [8.33,11.67]	12.00 [9.90,14.11]	11.00 [9.32,12.68]	-2.00 [-3.52,-0.47]
CycleINR→I3Net	5.87 [4.54,7.20]	7.39 [5.65,9.13]	6.63 [5.30,7.97]	-1.52 [-2.93,-0.11]
CycleINR→SAINT	6.49 [4.74,8.24]	8.30 [6.14,10.47]	7.40 [5.64,9.16]	-1.82 [-3.30,-0.33]
CycleINR→SwinIR	8.25 [6.20,10.29]	13.66 [11.20,16.11]	10.95 [8.92,12.98]	-5.41 [-7.04,-3.78]
HAT→CycleINR	9.11 [7.17,11.06]	13.19 [11.98,14.39]	11.15 [9.86,12.44]	-4.07 [-6.05,-2.10]
HAT→I3Net	9.25 [7.98,10.53]	10.09 [8.76,11.42]	9.67 [8.69,10.65]	-0.84 [-2.54,0.87]
HAT→SAINT	12.16 [10.61,13.71]	13.50 [11.88,15.12]	12.83 [11.49,14.16]	-1.34 [-2.96,0.28]
HAT→SwinIR	13.82 [12.37,15.27]	17.49 [15.78,19.21]	15.66 [14.28,17.03]	-3.67 [-5.12,-2.22]
I3Net→CycleINR	6.58 [5.16,8.00]	6.01 [4.52,7.49]	6.29 [5.32,7.27]	0.57 [-1.63,2.77]
I3Net→HAT	10.35 [8.97,11.72]	7.93 [6.30,9.55]	9.14 [7.93,10.34]	2.42 [0.68,4.16]
I3Net→SAINT	11.91 [10.52,13.30]	9.90 [8.84,10.95]	10.90 [9.91,11.90]	2.01 [0.57,3.46]
I3Net→SwinIR	7.35 [5.66,9.05]	8.01 [6.13,9.88]	7.68 [6.40,8.96]	-0.65 [-3.16,1.86]
SAINT→CycleINR	7.75 [5.93,9.56]	7.78 [6.19,9.36]	7.76 [6.48,9.04]	-0.03 [-2.29,2.24]
SAINT→HAT	12.94 [11.35,14.53]	12.46 [10.82,14.10]	12.70 [11.32,14.08]	0.48 [-1.08,2.04]
SAINT→I3Net	11.56 [10.26,12.86]	10.84 [9.58,12.09]	11.20 [10.16,12.24]	0.73 [-0.74,2.19]
SAINT→SwinIR	10.15 [8.37,11.92]	11.31 [9.30,13.33]	10.73 [9.17,12.28]	-1.17 [-3.26,0.93]
SwinIR→CycleINR	10.29 [7.56,13.01]	11.58 [10.09,13.07]	10.93 [9.14,12.73]	-1.29 [-3.84,1.26]
SwinIR→HAT	15.47 [13.82,17.12]	15.81 [14.26,17.36]	15.64 [14.29,16.99]	-0.34 [-1.98,1.30]
SwinIR→I3Net	8.28 [6.70,9.85]	8.90 [7.56,10.24]	8.59 [7.38,9.79]	-0.62 [-2.24,1.00]
SwinIR→SAINT	10.74 [8.92,12.57]	10.93 [9.05,12.80]	10.83 [9.26,12.41]	-0.18 [-1.99,1.63]

Table 41. ixi: all ordered-pair geometry contrasts, percentage points, with simultaneous 95% intervals per column. Mirror difference is an orientation diagnostic, not an observability-gain endpoint.

Ordered pair	Residue 1	Residue 3	Balanced	Mirror difference
CycleINR→HAT	20.07 [19.29,20.85]	22.17 [21.46,22.89]	21.12 [20.56,21.69]	-2.10 [-3.09,-1.11]
CycleINR→I3Net	12.34 [11.27,13.42]	13.92 [12.78,15.07]	13.13 [12.23,14.04]	-1.58 [-2.88,-0.28]
CycleINR→SAINT	22.46 [21.68,23.25]	22.64 [21.93,23.34]	22.55 [21.99,23.11]	-0.17 [-1.18,0.84]
CycleINR→SwinIR	21.57 [20.82,22.32]	23.15 [22.46,23.84]	22.36 [21.82,22.90]	-1.58 [-2.55,-0.61]
HAT→CycleINR	20.70 [19.85,21.54]	21.63 [20.83,22.43]	21.16 [20.56,21.76]	-0.93 [-2.08,0.21]
HAT→I3Net	18.28 [16.96,19.61]	18.39 [17.39,19.40]	18.34 [17.35,19.32]	-0.11 [-1.43,1.21]
HAT→SAINT	28.39 [27.31,29.48]	26.70 [25.96,27.43]	27.55 [26.85,28.25]	1.70 [0.45,2.94]
HAT→SwinIR	29.06 [28.00,30.11]	28.04 [27.25,28.84]	28.55 [27.85,29.25]	1.01 [-0.24,2.26]
I3Net→CycleINR	12.41 [11.14,13.68]	13.70 [12.40,15.00]	13.05 [12.08,14.03]	-1.29 [-2.99,0.41]
I3Net→HAT	18.21 [16.73,19.68]	17.76 [16.69,18.84]	17.99 [16.91,19.06]	0.44 [-1.02,1.91]
I3Net→SAINT	18.29 [16.78,19.80]	17.56 [16.43,18.69]	17.92 [16.83,19.02]	0.73 [-0.85,2.30]
I3Net→SwinIR	18.17 [16.70,19.65]	17.72 [16.62,18.83]	17.95 [16.88,19.02]	0.45 [-1.06,1.96]
SAINT→CycleINR	21.65 [20.81,22.49]	23.48 [22.68,24.27]	22.56 [21.97,23.16]	-1.83 [-2.97,-0.69]
SAINT→HAT	27.12 [26.08,28.17]	27.84 [27.05,28.63]	27.48 [26.79,28.18]	-0.72 [-1.98,0.54]
SAINT→I3Net	17.97 [16.65,19.30]	18.47 [17.41,19.54]	18.22 [17.22,19.22]	-0.50 [-1.85,0.86]
SAINT→SwinIR	27.99 [26.94,29.05]	28.24 [27.43,29.05]	28.12 [27.42,28.81]	-0.25 [-1.54,1.04]
SwinIR→CycleINR	22.01 [21.17,22.86]	22.76 [21.98,23.54]	22.39 [21.79,22.98]	-0.74 [-1.87,0.38]
SwinIR→HAT	28.64 [27.57,29.71]	28.36 [27.56,29.16]	28.50 [27.78,29.21]	0.28 [-0.97,1.53]
SwinIR→I3Net	18.25 [16.93,19.57]	18.29 [17.28,19.31]	18.27 [17.28,19.26]	-0.04 [-1.36,1.27]
SwinIR→SAINT	29.07 [27.95,30.18]	27.21 [26.47,27.95]	28.14 [27.42,28.86]	1.86 [0.59,3.12]

Table 42. Joint examination-bootstrap family summaries (pp). Intervals are descriptive percentile 95% intervals, not simultaneous across exclusions.

Cohort	Excluded architecture	Pairs	Mean [CI]	Median [CI]
fastmri	none	20	10.48 [9.63,11.35]	10.85 [9.86,11.74]
fastmri	hat	12	9.24 [8.37,10.14]	9.67 [8.75,10.39]
fastmri	swinir	12	9.77 [9.01,10.57]	10.23 [9.52,10.87]
fastmri	i3net	12	11.64 [10.67,12.65]	11.35 [10.19,12.45]
fastmri	saint	12	10.42 [9.59,11.29]	10.41 [9.50,11.21]
fastmri	cycleinr	12	11.32 [10.50,12.18]	10.85 [9.97,11.82]
ixi	none	20	21.77 [21.26,22.33]	21.77 [21.33,22.21]
ixi	hat	12	20.40 [19.86,20.99]	20.30 [19.77,20.88]
ixi	swinir	12	20.11 [19.56,20.70]	19.73 [19.19,20.31]
ixi	i3net	12	25.04 [24.59,25.51]	25.04 [24.59,25.53]
ixi	saint	12	20.24 [19.71,20.83]	19.73 [19.19,20.31]
ixi	cycleinr	12	23.09 [22.50,23.73]	22.91 [22.33,23.54]

6. Negative Results and Boundaries

6.1. Final-analysis boundaries

Every fastMRI fixed-reference ranking interval crosses zero; SAINT-to-I3Net has a negative point estimate. Four IXI fixed-reference intervals cross zero, all with I3Net as companion. The more favorable condition-specific paired-error endpoint is separately labeled and cannot replace fixed-reference inference. Near-orthogonal but no opposing neural sensitivities occur in the fixed-context sample, and individual approximation errors can exceed 100%. Acquired-only brain influence does not repair the old NingXia null/mixed result or establish a cross-architecture finite-response claim. The final score-search branch is closed.

6.2. One Acquired-Only Guard: Full Negative Evaluation

This experiment uses stored phase-0 predictions from two fixed pairs, HAT/SwinIR and I3Net/CycleINR. Each pair is evaluated on five target planes per examination. A target's scoring mask is the union of its two closest acquired phase-0 1% foregrounds. A single acquired nonzero-intensity Q99–Q1 range normalizes disagreement, error and acquired-anchor baselines. Both the mask and scale are computed without the missing target. Error evaluation then uses that same support. Model weights are dataset-specific historical checkpoints, not one cross-dataset model.

The 24 fastMRI case IDs are ordered by SHA-256 of a fixed prefix plus ID; the first 12 form development and the others internal evaluation. All 115 IXI cases are an additional historically reused transfer cohort. No patient appears in both new fastMRI parts, but historical analysis reuse prevents an untouched-holdout claim.

Only one candidate is tried: $G = \max(F_{\mathrm{dev}}(D_n), F_{\mathrm{dev}}(A_n))$, with right-continuous development empirical CDFs and acquired-range normalization. The dependency trigger is shared phase-0 evidence; no claim is made that the guard estimates neural influence. The thresholds are development Q75 of $\min(\mathrm{MAE}_a, \mathrm{MAE}_b)_n$ and Q25 of D_n. This requires both predictions to have high case-level error and low disagreement. Formula, CDF samples and thresholds are saved before the evaluation summary is computed. There is no threshold retuning or additional formula search.

The internal evaluation contains no positive cases for either pair; AUROC and AUPRC are undefined, not zero. IXI has two positives per pair. The guard's AP is .5087 and .0174 versus .5089 and .0144 for disagreement, and .5102 and .0181 for Anchor-CV. These point estimates are unstable with two positives and are not evidence of useful detection. No guard-minus-baseline interval excludes zero across the twelve comparisons (two pairs by six baselines) on IXI. Of 5,000 joint bootstrap draws, 3,918 retain both classes for both pairs; intervals are conditional on those nondegenerate draws and should not be treated as reliable rare-event coverage guarantees. The fastMRI family is undefined throughout. No further scoring family is searched.

Secondary error-rank correlations, AUROC, top-20% enrichment, all baseline scores, predictions and excluded bootstrap counts are in the machine-readable outputs. Acquired header spacing can be constant and is not a verified clinical spacing measure here. Missing or undefined metrics are retained as NA. The previous context-selection, jackknife and residualization failures below remain relevant; this candidate does not rescue a universal method claim.

Table 43. All candidates and comparators in the new guard audit. HS: HAT/SwinIR; IC: I3Net/CycleINR. Internal fastMRI evaluation has zero positives, so AP is undefined. IXI has two positives per pair; its AP point estimates do not establish superiority. ρ ranks normalized paired error.

Cohort	Score	HS AP	IC AP	HS ρ	IC ρ
fastmri	Guard	NA	NA	0.694	0.237
fastmri	Disagreement	NA	NA	0.811	0.462
fastmri	Anchor-CV	NA	NA	0.713	0.594
fastmri	Adjacent MAE	NA	NA	0.483	0.566
fastmri	Adjacent RMSE	NA	NA	0.434	0.217
fastmri	Acquired range	NA	NA	0.559	0.147
fastmri	Header spacing	NA	NA	NA	NA
ixi	Guard	0.509	0.017	0.150	0.138
ixi	Disagreement	0.509	0.014	0.305	0.016
ixi	Anchor-CV	0.510	0.018	-0.172	-0.128
ixi	Adjacent MAE	0.511	0.019	-0.131	-0.075
ixi	Adjacent RMSE	0.512	0.020	-0.152	-0.114
ixi	Acquired range	0.025	0.023	0.313	0.309
ixi	Header spacing	0.513	0.025	-0.191	-0.247

6.3. New-Architecture Influence Stability and Boundary

SAINT uses all 24 fastMRI examinations and three deterministic interior planes. I3Net/CycleINR use the first 24 geometry-eligible cases in frozen IXI test-list order and one central plane. Their regular four-anchor reference is $(-5, -1, +3, +7)$, with two overlap-three and five zero-overlap contexts. Zero-overlap influence is zero by physical-measurement definition and is excluded from the stability gate.

SAINT's broad influence associations have the expected direction for alignment, shared fraction, observable fraction, and disagreement. Its fixed-count estimates also have the expected signs, but only disagreement excludes zero. This supports a stable local influence signal in a non-transformer while falsifying a universal beyond-count advantage. I3Net/CycleINR remain valid phenomenon experiments because their controlled predictions do not depend on the finite-response stability classification.

6.4. Material Negative and Boundary Results

1. Integer overlap is not a monotone dose; the exact analytical ladder reverses at zero-to-one sharing.
2. Fixed-count continuous influence is null/mixed on NingXia; all eight-patient intervals cross zero.
3. Local response rankings are stable to tested perturbation parameters but not invariant across physical planes.
4. The local cancellation chain is stronger than the global context-ranking chain; no causal mediation claim is supported.
5. Anchor-CV is not uniquely informative or universally superior. Adjacent differences are statistically similar and can be stronger; output PC wins several IXI comparisons.
6. Examination scores do not localize the worst plane and are not calibrated pixelwise uncertainty.
7. Densification results are retrospective; the same-model audit retains a negative NingXia Res-SRDiff boundary and omits scanner-time, motion, and clinical utility.
8. Minimum-influence context selection transfers below random/PC on IXI and is heterogeneous on NingXia.
9. Multi-anchor jackknife IQR collapses from development to confirmation/transfer and is far below adjacent MAE.
10. Covariance residualization gives negligible IXI gain with intervals crossing zero and mixed/tied results elsewhere.

Items 8–10 test abandoned mechanism-derived methods. They are not main-paper contributions, but are retained here because they prevent the reader from inferring that the mechanism already yields an improved universal score.

7. External Evidence and Its Limits

7.1. No new untouched-cohort confirmation

No locally authenticated unused cohort was available under the optional final branch. Existing TopCoW/CAS analyses below remain analytical studies on public challenge training collections with the documented outcome-informed history, not newly untouched external neural confirmation.

7.2. External Support, Scale, and Anatomical Checks

These external MRA experiments are exploratory. The external MRA protocol was written before outcome calculation; it uses a different cohort from the historical brain T2 studies. All native images and labels remain unchanged. Complete cases, protocol versions, 250 source hashes, all mask choices and fixed-estimator benefit results are retained in the final

Table 44. historical architecture-family influence audit. Stability entries are rank correlations with the $+1\%$ primary statistic. A model is claim eligible only if all are at least .90.

Model	$+.5\%$	$+2\%$	-1%	Broad $\rho(I$,alignment)	$\lvert\rho_I\rvert - \lvert\rho_{count}\rvert$ [CI]	Fixed-count partial ρ [CI]
SAINT	.940	.964	.995	.706 [.680,.734]	.014 [−.002,.029]	.063 [−.021,.149]
I3Net	.859	.737	.939	.776†	−.006†	.434†
CycleINR	.779	.857	.876	.789†	.007†	.229†

†Exploratory only after the documented stability gate failed; excluded from influence claims.

reproducibility and evidence map.

Correcting bank-score provenance. The archived `target_0.01` field denotes the target model's prediction mask, not ground truth. We independently recompute 84 mask/target/partner combinations from original predictions on the lexicographically first case in each cohort; maximum difference is 1.42×10^{-14}. Linear/Cubic are candidate members; the smaller denominator comes from neural-only target precomputation. The historical architecture studies's intervention masks remain ground-truth-derived. Across all 506 eligible model–case rows, acquired-input masks give median benefit correlations .863/.723, prediction masks .835/.710, and bank unions .805/.519 on fastMRI/NingXia. Acquired-mask partial medians given current error are $-.058/.425$. No mask is chosen based on these outcomes.

Crossing score and endpoint units. For the 115 recovered IXI I3Net pairs, use the same acquired-anchor 1st–99th percentile range r to form all four contracts: (s, b), $(s/r, b)$, $(s, b/r)$, and $(s/r, b/r)$. Anchor-CV correlations are .843, −.238, −.375, and .539; adjacent MAE gives .845, −.236, −.373, and .551. Bootstrap count is 10,000 and seed 20260909. At 50% selection, matched normalized Anchor-CV exceeds random by .000655 normalized MAE units [.000151,.001537], while the same score against raw benefit remains below random. Shared scaling can induce correlation, so unit consistency is not a causal or calibration guarantee.

External MRA protocol. We use all 125 locally available annotated TopCoW MRA cases (public challenge training data, no fitting in this experiment), with 10779 eligible odd planes. L is linear interpolation; C is four-point Lagrange interpolation with no output clipping. For each phase, all four cubic anchors must lie inside the image; neither boundary duplication nor label-based plane selection is allowed. L_0–C_0 and C_0–L_0 are shared controls; their disjoint partners are C_2 and L_2. The input-support mask is the union of the two nearest acquired phase-0 anchor masks, each thresholded at 1% of that anchor's maximum absolute intensity. The full-image and target-derived-mask alternatives are sensitivity contracts, not pooled replicas. Within each case, means are first taken within target residues 1 and 3, then the two residue means are equally weighted. Cases are the bootstrap units; 10,000 common draws are used. Local-gain simultaneous intervals use the maximum standardized deviation across the two estimator pairings within each mask. Other intervals are exploratory pointwise.

Vessel labels define region-restricted intensity error only. Empty-vessel planes are recorded as undefined, not zero; every case retains its actual vessel-plane denominator. The ranking endpoint is the fixed phase-0 target estimator's error, so it does not change between the shared and disjoint scores. The $\times 2/\times 4$ benefit comparison uses identical estimator rules and common odd planes. Raw and acquired-range-normalized endpoints/scores, both estimators, and fixed 20%/50% budgets are all retained.

Legacy split and geometry audit. The old fastMRI split contains one patient in both train and validation, but test patients overlap neither. Historical checkpoint-selection bias cannot be quantified from the existing outputs. For all 24 paper test cases, original DICOM headers and first/middle/last pixel planes confirm ordering, parallel directions, and monotone physical positions. The stored diagonal NIfTI affine does not retain patient orientation, so agreement of inherited affine matrices alone is insufficient evidence. Source documents in other projects are used for traceability, without importing their developed-model performance as independent evidence.

Anatomical weighting: development and fixed-rule transfer. After the TopCoW baseline analysis, we fix a bright-anchor mask at the 95th percentile within each acquired anchor's 1%-foreground and unite the nearest two masks. The threshold is never searched. TopCoW reuses the development cohort; CAS2023 receives this rule unchanged on all 100 image/label pairs and 6,412 eligible planes. Image and label shapes/affines match. No file hashes match TopCoW, which does not prove person-level nonoverlap without linked identifiers. Unknown CAS header provenance precludes a physical-spacing claim. Cases, residue balancing, estimator rules, and bootstrap count/seed match the preceding protocol.

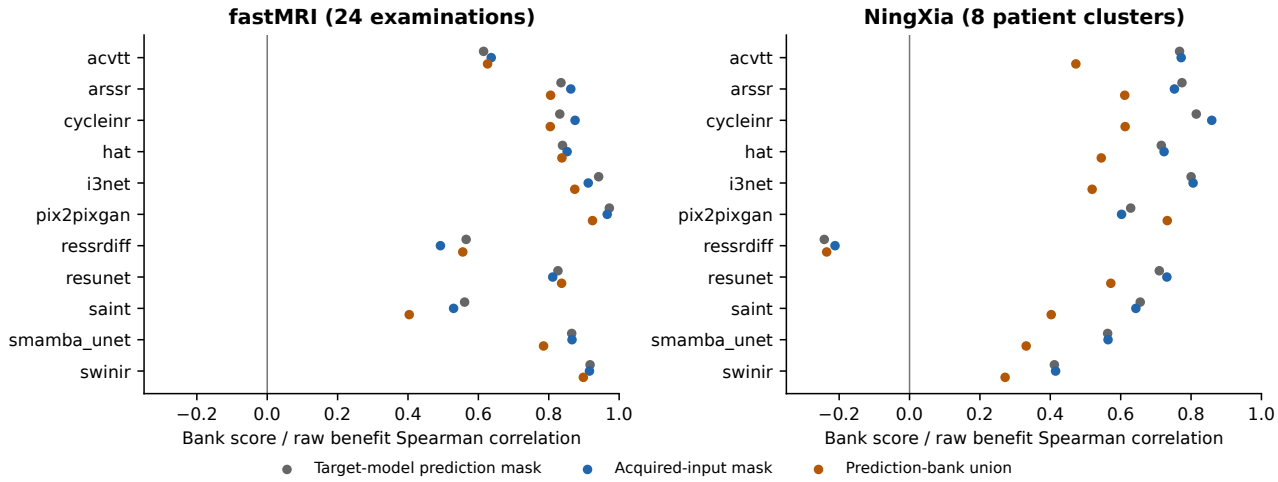


Figure 11. All 11 neural target models under three score-support contracts. Points are raw benefit correlations, not confidence limits. Linear/Cubic remain candidate members; neither was precomputed as a target model. All model-specific bootstrap intervals and paired mask differences are retained in the numerical record.

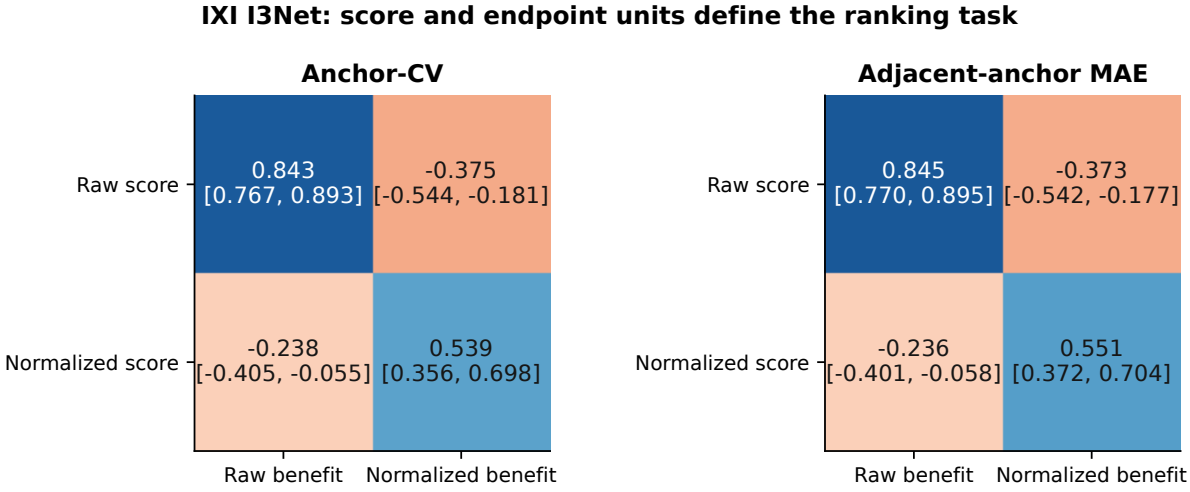


Figure 12. Score–endpoint unit contracts on IXI. Each cell gives Spearman correlation and an exploratory pointwise 95% examination-bootstrap interval. Both favorable diagonal cells and adverse off-diagonal cells are reported.

Before bright weighting, changing shared to disjoint contexts worsens TopCoW vessel-risk coverage: normalized excess risk area increases by .00140 [.00028,.03211] for L and .00135 [.00026,.03120] for C. Full-image score support also yields a negative vessel-ranking change for L, $-.0490$ $[-.1016, -.0003]$. These qualified negative sensitivity results remain part of the evidence. On TopCoW, all four vessel-risk coverage changes favor bright weighting and all four general-error rankings deteriorate. On CAS2023, the disjoint vessel-risk coverage changes are $-.01047$ $[-.02389, .00034]$ and $-.01074$ $[-.02482, .00089]$; shared changes also cross zero. Negative changes in normalized excess risk area favor the new score. Both contexts lose general-error ranking on CAS2023. This is a target-specific ranking transfer with unresolved screening benefit, not a universal score improvement. The full numerical record includes all endpoints and contexts, including nonsignificant and adverse results.

Table 45. Bright-anchor minus general-support vessel-error ranking. Intervals are simultaneous across L/C within each cohort/context. TopCoW is exploratory rule development; CAS2023 is fixed-rule transfer. General-error changes are retained to expose the tradeoff.

Cohort	Context/target	Baseline ρ	Bright ρ	$\Delta\rho$ [95% CI]	General $\Delta\rho$
TopCoW	shared/L	0.531	0.633	0.102 [0.008,0.195]	-0.216
TopCoW	shared/C	0.509	0.613	0.104 [0.011,0.197]	-0.217
TopCoW	disjoint/L	0.488	0.705	0.217 [0.100,0.334]	-0.250
TopCoW	disjoint/C	0.471	0.693	0.222 [0.110,0.334]	-0.235
CAS2023	shared/L	0.908	0.918	0.010 [-0.038,0.059]	-0.136
CAS2023	shared/C	0.903	0.908	0.005 [-0.048,0.057]	-0.138
CAS2023	disjoint/L	0.880	0.953	0.073 [0.017,0.130]	-0.085
CAS2023	disjoint/C	0.877	0.947	0.070 [0.014,0.126]	-0.084

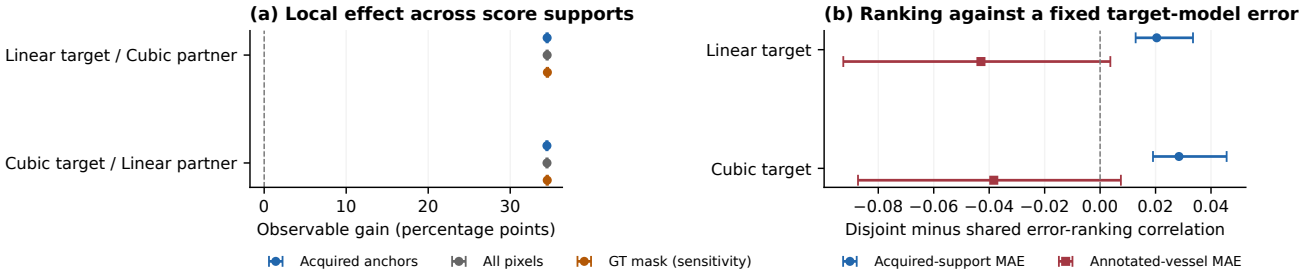


Figure 13. External MRA local geometry and anatomical ranking. The analytical experiment broadens contrast and endpoint coverage, but does not replicate the five neural models on MRA. Local error exposure is distinct from useful global or vessel-error ordering.

8. Reproducibility and Evidence Map

8.1. Executable reproduction and report cards

Run `python reproduce_main.py --complete` in `FINAL_SUBMISSION_PREPARATION/`. The CPU-only path reads preserved target, prediction and finite-response arrays; reconstructs the historical and primary observation tables and clustered statistics; independently remeasures the support and geometry sensitivities; and regenerates the main figures, tables and report cards. Historical summaries are assertion targets, not numerical inputs. Historical observation keys still define the frozen selections. No training or network inference is required. Array bindings, differences and tolerances are recorded in `REPRODUCTION_CHECK.md`; environment, commands and detailed legacy paths are in `REPRODUCIBILITY.md`.

The report card separates local observability, the named ranking endpoint, beyond-count influence, fixed-count association, stability, information access and claim eligibility. Unmeasured endpoints are NA. Cross-architecture phase rows do not inherit within-model response evidence, and probe rows do not inherit phase-ranking effects. Eligibility refers to the named claim under its contract, not individual reconstruction correctness.

8.2. Frozen evidence and post-hoc patch

The accepted primary studies remain under `FINAL_SCIENTIFIC_STRENGTHENING/`. The submission patch separately preserves the accepted manuscript and final tables and hashes all array inputs before sensitivity calculation. Its `FINAL_SUBMISSION_PATCH_PROTOCOL.md` distinguishes frozen primary evidence, new post-hoc sensitivities and editorial changes; `freeze.json` binds the protocol and manifests.

The three complete sensitivity files have stems `CROSS_ARCH_SUPPORT_SENSITIVITY`, `CROSS_ARCH_GEOMETRY_SENSITIVITY` and `CROSS_ARCH_FAMILY_ROBUSTNESS`. They retain every ordered pair, family summary and unavailable geometry contract. The acquired-union rule is the existing foreground-union protocol; mirror comparison follows the historical reflection-class analysis while retaining the new study's mean estimand. Independent pixel/geometry reconstruction is distinguished from reuse of the frozen bootstrap aggregation code.

8.3. Provenance, negative results and access boundary

The companion claim matrix maps every main claim to its unit, array source, uncertainty contract and figure/table. The research ledger retains historical clean controls, analytical ladders, response/stability audits, recovered-model-bank and budget results, densification and external MRA studies, and every qualified adverse result. Detailed workspace paths and protocol amendments are retained there rather than repeated in the scientific narrative.

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